

The Impact of Government Funding on Domestic Violence Shelters and Domestic Violence Incidents

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Abstract

Domestic violence (DV) shelters receive millions of dollars in government funding each year, yet no empirical evidence exists on how these funds affect shelter operations or community-level violence. I construct a novel panel dataset of DV shelters in Georgia from 2013 to 2023, combining IRS Form 990 filings and government grant records for 46 shelters, and link this panel to county-level measures of domestic violence and child maltreatment. Using two-way fixed effects models with contemporaneous and lagged grant funding, I find that a \$100,000 increase in government grants raises shelter revenue by \$61,900. Private donations decline by \$38,900 per \$100,000 in grants, a 13 percent drop relative to the sample average. Total expenditure rises by \$57,300 per \$100,000 in grants, directed primarily toward compensation and program services. Employee counts also rise with increased grant funding. Estimated effects on community-level violence and child maltreatment are small and inconsistent across specifications. Of the outcomes I examine, only one survives correction for multiple testing: a small increase in female assault hospital discharges in the year following a grant increase. I find no evidence that funding changes service utilization at shelters. These results indicate that government grants increase shelter revenue, expenditure, and staffing, while county-level violence and child maltreatment outcomes show little detectable change.

1 Introduction

Domestic violence (DV) is a prevalent public health concern in the United States where almost half of women and four in ten men report experiencing DV in their lifetime (Leemis et al., 2022). DV shelters offer resources for victims attempting to leave abusive situations such as safe housing, case management, and counseling. These organizations rely on government funding, specifically through the Victims of Crime Act (VOCA). In Georgia alone, the federal government distributed \$40.9 million through VOCA funds in 2025. Yet VOCA funding has fallen sharply, from \$1.9 billion nationally in 2023 to \$1.3 billion in 2024 (Office for Victims of Crime, 2026), and we know little about how these funds are used and how they affect shelter operations and domestic violence outcomes.

The lack of research in this area reflects a fundamental data challenge. No publicly available, longitudinal, comprehensive database of DV shelters exists. While some government agencies publicly disclose grant awardees in certain years, there is no consistent, comprehensive record of which shelters operate from year to year, let alone their funds received or expenditures. I overcome this challenge by scraping IRS Form 990 filings and constructing a novel panel dataset from 2013–2023 covering 47 DV shelters in Georgia. The shelter-level database includes information about finances, operations, and grant funding. I link this shelter-level data to county-level DV measures, enabling the first empirical analysis of how government funding shapes shelter operations and community-level violence.

I estimate two-way fixed effects (TWFE) models that exploit within-shelter variation in government funding, allowing both contemporaneous and one-year lagged effects. I find that DV shelters respond substantially to changes in government funding, with grants translating into increased revenue, expenditures, and staffing. A \$100,000 increase in government grants raises shelter revenue by \$61,900. The results suggest crowd-out of private donations with an average decrease of \$38,900 per \$100,000 increase in government grants, which is around a 13% decline relative to the annual average \$304,600. Total expenditure rises by \$57,300 per \$100,000 increase in grants, concentrated on compensation and program

services. These results demonstrate that DV shelters translate government funding directly into organizational capacity. I next consider whether this expanded capacity is detectable in county-level violence outcomes such as female homicide and assault rates. Estimated effects on community-level violence and child maltreatment are small and sensitive to specification. The only outcome that shows a robust association with grant funding is a small increase in female assault hospital discharges in the year following a grant increase. Limited observable effects on community-level outcomes may reflect a lack of detectable impact, or effects operating through channels this analysis is not designed to capture, such as service quality or other measures of client well-being.

My results contribute to three distinct literatures related to non-profit organizations and DV. First, my paper contributes to a broader literature on government funding and nonprofit organizations, which has established that public grants can crowd out private donations, largely through reduced fundraising effort rather than donor substitution (Andreoni & Payne, 2011; Froelich, 1999; Pizzini & Sterin, 2023). I extend this literature by studying domestic violence shelters. Funding stability is a first-order concern for organizations that depend on it: staffing levels and retention hinge on the ability to sustain revenue from year to year. It is also consequential for the populations these organizations serve: prior work links funding stability to program outcomes in other nonprofit sectors (Kim, 2017). I provide the first evidence on how government grants shape the finances and operations of DV shelters.

A large literature examines the effect of criminal justice interventions on domestic violence, including arrest policies, prosecution, restraining orders, and police presence, with mixed results (Aizer & Dal Bó, 2009; Black et al., 2023; Dugan et al., 2003; Raissian, 2016). These interventions primarily alter the abuser’s incentives through deterrence or legal consequence. A growing body of work also examines non-criminal levers: Sims et al. (2024) find no significant effect of state EITC expansions on lethal intimate partner violence, Carr and Packham (2021) show that SNAP disbursement timing affects DV incidence, and Geddes and Holz (2026) find that rent control reduces hospitalizations for female assault. These

interventions ease the economic or housing constraints that shape household bargaining and exposure to violence. DV shelters operate outside the criminal justice system, expanding victims’ safety and exit options directly through refuge, safety planning, and advocacy. I contribute to both literatures by providing the first evidence on whether funding for these shelter programs affects violence and child maltreatment outcomes.

The most closely related work examines the effect of DV shelter access on homicides. Schechter (2023) finds that opening a shelter in counties with no prior access reduces intimate partner homicides with female victims by approximately one homicide per 100,000 people every three to four years. Sims (2021) finds no causal effect of changes in shelter capacity on IPV or DV homicides, with small, precisely estimated effects, and offers suggestive evidence that service quality may matter more than bed count.¹ Both papers rely on variation at the extensive margin through shelter openings, closings, and physical expansions, which are relatively rare events.² By contrast, I exploit variation in shelter funding and consider additional outcomes including shelter operations, DV-related hospitalizations, and child-maltreatment.

The remainder of the paper proceeds as follows. Section 2 describes the domestic violence shelter sector and the federal funding programs that finance it. Section 3 details the construction of the panel dataset. Section 4 presents the empirical strategy. Section 5 reports results, beginning with grant effects on shelter finances and operations before turning to violence outcomes and utilization. Section 6 presents a series of robustness and sensitivity analyses. Section 7 concludes.

¹Sims (2021) is not publicly available; characterization is based on discussion with the author.

²For example, there are no DV shelter openings or closings in GA during the 2013–2023 sample period.

2 Background

2.1 Domestic Violence Shelters

Domestic violence shelters provide immediate housing and structured programs for victims fleeing abusive relationships and households. Unlike homeless shelters, DV shelters provide residents a private room for the duration of their stay. They are also considerably smaller: Georgia’s DV shelters report approximately 1,000 beds,³ compared to roughly 5,300 beds across the state’s homeless shelters (Department of Housing and Urban Development, 2024). Residents have access to a range of services that can include counseling, childcare, legal aid, and housing assistance (Craig, 2023), with shelters in Georgia reporting as many as 43 distinct services (Criminal Justice Coordinating Council, 2017–2022). Staffing structure determines the breadth of services shelters can offer: shelters are typically organized around a program director and a set of specialized advocates — covering areas such as legal assistance, housing, and children’s services — whose caseloads determine the services victims can access.⁴ Bed capacity is also a constraint: on a single day in September 2024, 1,393 individuals were in program-provided housing in Georgia while 275 housing requests went unmet (National Network to End Domestic Violence, 2024).

Individuals who seek shelter services often have few viable alternatives for safe housing. Adams et al. (2008) found that 102 out of 103 women who were receiving services from an undisclosed domestic violence organization had experienced economic abuse. Separately, Adams et al. (2012) find that intimate partner violence reduces job stability and economic well-being among low-income women, with effects persisting up to three years after the violence ends. Abusers further limit victims’ alternatives by systematically isolating them from family and friends (DomesticShelters.org, 2015), and 73 percent of victims report remaining in abusive situations due to financial concerns (Adams et al., 2020). Between 22 and 57

³Bed count was measured through reports and the author’s contact with shelters.

⁴The description of staffing structure is based on staff pages from Georgia DV shelter websites and staffing constraint reports from the National Network to End Domestic Violence (National Network to End Domestic Violence, 2024).

percent of homeless women identify domestic violence as the immediate cause of their homelessness (Center for Impact Research, 2004; Institute for Children & Poverty, 2004; Wilder Research Center, 2004).

Understanding how government funding affects shelter staffing and service delivery is therefore important for victim well-being. Changes in funding may affect whether specialized advocates can be retained, whether legal aid partnerships can be sustained, and the services that can be offered. Although the ultimate policy objective is to reduce violence and abuse, shelter finances and operations are important outcomes in their own right because they represent the organizational responses through which funding reaches victims. Whether these organizational responses result in changes in community-level violence, however, depends on several distinct channels linking funding to victim outcomes.

First, additional funding may improve the quality of services provided to victims who already receive help from shelters. Additional funding may allow shelters to provide more intensive counseling, more consistent case management, and expanded legal advocacy to clients they already serve. These improvements could materially benefit shelter clients without producing any detectable change in community-level violence, since they affect victims already in contact with shelter services rather than the broader population of victims in the county.

Second, additional funding may increase the capacity of shelters to serve additional victims. If shelters are capacity constrained, additional funding could be used to hire more caseworkers or add beds, extending services to victims who would not otherwise receive help. Extending services to additional victims could reduce community-level violence by helping more victims leave abusive relationships or access protective resources. Distinguishing this channel is complicated, however, by measurement: bringing more victims into contact with shelters, hospitals, or the legal system could increase the number of cases that appear in administrative records such as hospital discharge data, even if underlying violence declines. An increase in measured violence following capacity expansion is therefore consistent with

either a rise in underlying violence or an increase in reporting among victims newly connected to services.

Third, additional funding may expand outreach without expanding capacity. If additional funding increases hotline availability, advocacy, or public awareness efforts without a corresponding increase in shelter beds or staff, victims may attempt to leave abusive relationships without access to adequate housing or safety resources. Leaving an abusive partner is widely recognized as one of the most dangerous periods for victims of intimate partner violence (Domestic Violence Coordinating Council, 2011). Outreach that increases attempts to leave without expanding capacity could therefore increase community-level violence.

These three channels imply different, and at times opposing, predictions for the effect of shelter funding on community-level violence. Consequently, a small or null estimated effect on violence outcomes does not indicate that organizational improvements failed to reach victims. I therefore treat the downstream violence outcomes as a secondary and imprecise measure of organizational impact, rather than as the primary expected consequence of increased shelter funding.

2.2 Victims of Crime Act Funding

Government funding is a key resource for domestic violence shelters — on average, 68 percent of shelter revenue in Georgia comes from government funding.⁵ The largest funding source is the Crime Victims Fund (CVF), established by the Victims of Crime Act of 1984 (Office for Victims of Crime, 2026). The CVF is financed by fines and penalties from federal criminal cases. Congress limits the amount distributed to states through grants each year. From its introduction in 2000 through 2014, the cap ranged from \$500 million to \$700 million. In 2015, Congress tripled the cap, and annual distributions have ranged from \$2.4 to \$4.4 billion in the years since. After 2019, the CVF balance began to decline, partially eroding the post-2015 gains (Office for Victims of Crime, 2026).

⁵Author’s calculations from the DV Shelter Database (Kearns, 2026).

States receive allocations proportional to their share of the national population. Georgia, with roughly 3 percent of the U.S. population, receives approximately 3 percent of total CVF funds distributed to states. Federal rules require states to direct at least 10 percent of CVF-funded Victim Assistance Grant funds specifically to domestic violence programs (Office for Victims of Crime, 2026); in practice, Georgia has consistently allocated around 30 percent. Within Georgia, the Criminal Justice Coordinating Council (CJCC) is responsible for distributing federal funds to individual shelters. The CJCC’s allocation criteria are not publicly documented. In a November 2025 interview, a director at the Georgia Coalition Against Domestic Violence (GCADV), which coordinates between CJCC and shelters on VOCA funding applications, reported that the GCADV had helped CJCC develop a standardized allocation rubric.⁶ When I contacted CJCC directly to request a copy of the rubric, CJCC staff confirmed that it had been developed but never implemented, and did not describe any alternative systematic process used to allocate funds across shelters.⁷ Thus, funding decisions do not appear to be governed by a publicly-documented allocation formula, making future funding difficult for shelters to predict.

The 2015 cap increase represented a sustained funding shift, and distributions rose through 2018 before the fund balance began to decline. Shelters experienced different funding trajectories depending on CJCC allocation decisions, generating the within-shelter variation in government grant funding that the empirical analysis exploits.

3 Data Description

3.1 Domestic Violence Shelter Database

The primary limitation to studying DV shelters is data. No publicly available, longitudinal, comprehensive database of DV shelters exists.⁸ The administrative alternatives are limited:

⁶Author interview with a director at the Georgia Coalition Against Domestic Violence, November 2025.

⁷Author correspondence with CJCC staff, November 2025.

⁸The absence of a comprehensive database reflects two features of the sector: many shelters avoid public disclosure of their location and operations for victim safety reasons, and while some organizations track

the Criminal Justice Coordinating Council tracks only shelters receiving VOCA grants over a restricted set of years,⁹ and County Business Patterns (CBP) data aggregates DV shelters together with family, youth, and homeless shelters, making it impossible to isolate DV shelters specifically.¹⁰

DV shelters are nonprofit organizations and are required to file an annual Internal Revenue Service (IRS) Form 990, which includes the organization’s mission statement, top three program descriptions, and detailed financial information. Throughout this paper, I use “DV shelter” to refer to the filing organization, since Form 990 data are reported at the organization level rather than the program level. For most organizations in the database, the shelter and its related services, such as housing, legal advocacy, counseling, and hotline operations, constitute the organization’s core mission. For a small number of organizations, however, the shelter operates as one program within a larger nonprofit whose other lines of business are unrelated to domestic violence services. Because nonprofits with gross receipts above the IRS filing threshold must file annually, I use these filings to construct a novel dataset of DV shelters in Georgia from 2013–2023, cross-validated against multiple independent sources (Kearns, 2026). I identify 47 of the 48 DV shelters operating in Georgia, with continuous annual observations and detailed financial information unavailable in either alternative source. I build the panel in three steps.

First, I start with the universe of tax-exempt filers, roughly one million organizations, and filter to those classified as providing human or health services. I then use Optical Character Recognition (OCR) to extract mission and program statements from the two most recent 990 filings for each organization.¹¹ Any organization mentioning DV-related

shelters, these records are typically point-in-time snapshots without longitudinal coverage or systematic validation of whether listed organizations operate a shelter.

⁹There are four DV shelters that receive no government grants in at least one year during the sample period.

¹⁰The CBP data shows that four counties in Georgia open their first shelter during the sample period. However, two of these counties had a DV shelter already operating according to the 990 data. County Business Patterns data is drawn from Census Bureau (2023) and compared against the DV Shelter Database for Georgia over the 2013–2023 sample period.

¹¹Later filings are more likely to be e-filed and therefore have higher OCR accuracy. Only two years of filings were used for computational feasibility. Organizations serving DV victims typically retain DV-

terms is flagged as a potential DV service provider. These terms were derived from a review of 990 filings from shelters identified through an independent online search and are reported in Appendix A.

I use a Large Language Model (LLM) to extract structured data from the 990 filings for each flagged organization. An LLM is a machine learning model trained on large amounts of text that can be instructed to read a document and extract specific fields in a structured format, in this case financial and operational variables from standardized IRS form entries. The model was trained on 30 Form 990 filings per field type, such as total revenue, program service expenses, compensation of officers, and mission statement text. Additionally, I trained the model across the range of 990 form variants so that it could accommodate variation in form layout and reporting practices, and the model was validated out-of-sample before deployment. The LLM generates an initial extraction of the address, programming, and financial variables described in Appendix B. The address recorded on the 990 is the organization’s business address, which for shelters is typically a PO box rather than the physical shelter location, consistent with standard safety practices of concealing shelter addresses from public records. I assign each shelter to the county of its recorded business address.¹² All extracted records were then manually reviewed, with missing or unclear entries corrected by hand.

Finally, I cross-validate the resulting list of shelters against point-in-time snapshots from the CJCC, DomesticShelter.org, and the GCADV. I examine prior versions of these lists using the Internet Archive’s Wayback Machine and search archived news reports to identify when organizations began or ended shelter services. Through this process, I identify 48 DV shelters operating in Georgia during the sample period. One shelter, a large national nonprofit whose parent organization files a consolidated Form 990 that does not

specific language in their mission statements over time, since serving this population is central to, rather than incidental to, these organizations’ missions. Any DV service providers whose two most recent filings do not use these terms are still recoverable through the cross-validation and archival review described below.

¹²Where shelter websites report the county of physical operation, this county matches the county of the recorded 990 address in every case.

report program-level financial information, cannot be identified in the Form 990 data and is therefore excluded from the database entirely. Another shelter is identifiable in the Form 990 data but operates as a small domestic violence program within a large, multi-service nonprofit whose finances are dominated by other activities. Because its financial statements are not comparable to those of stand-alone domestic violence shelters, it is excluded from the financial analysis. The final analysis panel therefore includes 46 shelters.¹³ I observe no shelter openings or closures during the sample period.

Government grants are allocated on the federal fiscal year, which begins October 1st. Because shelters define their own fiscal years, I assign each shelter’s 990 filing to the federal grant award year using the October 1st start date. For example, a shelter filing a 990 for the fiscal year June 2013 to June 2014 is assigned to award year 2013, capturing the October 1st, 2013 grant allocation. Therefore, the treatment variable reflects the federal funding cycle driving variation in the data.¹⁴

3.2 CJCC Shelter Utilization and Services Records

The CJCC, the state agency responsible for distributing federal grant funds to DV shelters in Georgia, requires all grant recipients to report program statistics on a quarterly basis. These reports provide a rare window into shelter operations that is unavailable in the 990 data (Criminal Justice Coordinating Council, 2017–2022). I obtained annual shelter-level records through a direct data request to the CJCC, which aggregated the underlying quarterly reports before sharing them. The data cover 46 of the 47 shelters in the 990 database. One shelter operates without government funding and is therefore not a CJCC subgrantee.¹⁵ The data contain victim counts beginning in 2017, with a breakdown by service type beginning in 2018 and ending in 2022. Shelters report up to 40 distinct service categories, such as

¹³The DV shelter database is being expanded to the national level.

¹⁴The variation in fiscal year definitions across shelters raises the concern that the assignment of 990 filings to federal award years may introduce measurement error in the treatment variable. Restricting the sample to shelters operating on a December fiscal year, which comprises the majority of shelters in the panel, does not qualitatively change the financial results. Full results are presented in Section 6.

¹⁵This shelter is included in all 990-based financial analyses but excluded from the utilization analysis.

emergency housing, case management, and legal aid. I examine shelter service provision using three complementary measures: the total number of people served in each shelter-year, the number of people receiving each type of service, and an indicator for whether each service is offered.

The shelter utilization data have two limitations. First, victim counts may include duplicates because an individual who receives multiple services or seeks services from more than one shelter in the same year is counted multiple times. This is unlikely to affect the analysis, which focuses on shelter-level service provision rather than the number of unique individuals served statewide, unless funding changes systematically alter the extent to which survivors use multiple shelters. Second, the utilization data are available only for 2017–2022. As a result, the utilization analysis is based on substantially less funding variation than the financial and violence analyses.

Grant funding and service offerings show too little within-shelter variation during 2018–2022, the period for which service-level data are available, to support causal estimates of grant effects on service provision. I use the CJCC service records descriptively in Section 3.7 instead, to characterize the range of services Georgia shelters provide during this period.

3.3 Uniform Crime Reporting Supplemental Homicide Reports

DV homicide data come from the FBI’s Uniform Crime Reporting Supplemental Homicide Reports (UCR-SHR), which cover all homicides reported to law enforcement in Georgia from 2013 to 2019 (Federal Bureau of Investigation, 2013–2019). The sample ends in 2019 to avoid contamination from the sharp and well-documented rise in homicides during the COVID-19 pandemic. Stay-at-home orders are documented to have increased domestic violence incidents by confining victims with abusers for extended periods (Leslie & Wilson, 2020). The same period also coincided with a sharp, well-documented decline in CVF deposits, which reduced VOCA-funded victim assistance grants nationally (Office for Victims of Crime, 2026). Pandemic-era variation in both violence and funding would therefore confound the

effects of interest in a way that ending the sample in 2019 avoids.

The UCR-SHR includes information on victim and perpetrator characteristics and their relationship, which allows me to identify DV-related incidents. I construct four measures of lethal DV. DV homicides are defined as incidents where the perpetrator is a sibling, parent, grandparent, child, or current or past intimate partner. A subset, IPV homicides, restricts to incidents where the perpetrator is a current or past intimate partner. Lethal violence is a rare outcome. It is widely studied in this literature because it has lower rates of under-reporting than non-lethal DV. But under-reporting and limited statistical power remain concerns even for this measure. DV and IPV homicide measures rely on accurate reporting of the perpetrator’s relationship to the victim. Relationship information is sometimes missing or unknown in UCR-SHR records, which could lead to undercounts of true DV and IPV homicides. I construct an additional measure that does not rely on relationship coding. I count homicides where the victim is female between the ages of 18 and 34, the age range with the highest rates of intimate partner violence (Black et al., 2011). I similarly count homicides where the victim is under 18, to measure lethal violence against children without relying on relationship coding to classify the incident as domestic violence. DV and IPV homicides are scaled per 100,000 total county population, female homicides per 100,000 female county population ages 18–34, and child homicides per 100,000 county population under 18.

3.4 HCUP Emergency Department and Inpatient Discharge Records

While UCR homicides reflect the most extreme DV, I use the Healthcare Cost and Utilization Project (HCUP) Emergency Department and Inpatient Discharge Records (Healthcare Cost and Utilization Project, 2013–2019) to examine less severe but more prevalent forms of violence. The data consist of the universe of discharges for Georgia hospitals from 2013 to 2019, with observations at the discharge record level. For each discharge, I observe the year, patient zip code of residence, sex, age, and up to 25 diagnostic codes.

I construct three measures of DV-related hospital utilization. The first two measures, IPV and DV discharges use cause-of-injury diagnostic codes that specifically identify abuse, including physical, sexual, psychological, and financial abuse.¹⁶ The third measure counts assault discharges for female patients ages 18–34 regardless of relationship coding (Aizer, 2010; Geddes & Holz, 2026) and is therefore less sensitive to variation in how abuse is classified across hospitals or over time. Abuse-coded discharges can reflect either underlying violence or help-seeking behavior. This ambiguity is itself informative rather than a data limitation: if shelter funding encourages more victims to seek medical care, that behavioral response is a meaningful margin of interest. All measures are aggregated to the county-fiscal-year level using patient zip code of residence.¹⁷ DV and IPV discharge counts are scaled per 100,000 total county population. Female assault discharges are scaled per 100,000 female county population ages 18–34.

3.5 National Child Abuse and Neglect Data System

Data on child maltreatment come from the National Child Abuse and Neglect Data System (NCANDS), which contains the universe of child abuse and neglect reports submitted to state child protective services agencies (U.S. Department of Health and Human Services, 2013–2019). I use Georgia data from 2013–2019, which provide complete statewide county coverage. I aggregate reports to the county-fiscal-year level using the county to which each report was first assigned for investigation, which corresponds to the child’s county of residence.¹⁸

I construct three measures from the NCANDS data. The first is the count of substan-

¹⁶The diagnostic codes follow ICD-9 classification through the third quarter of 2015 and ICD-10 thereafter. Intimate partner violence (IPV) refers exclusively to violence involving current or former intimate partners, whereas domestic violence (DV) also includes violence involving other family or household members. The abuse codes are not mutually exclusive.

¹⁷Patient zip code ensures that discharge records are attributed to the locality of the victim rather than the locality of the treating facility.

¹⁸Georgia maintains a Division of Family and Children Services (DFCS) office in every county, though nine offices serve two counties from a single location. For these offices, the county of assignment may not always correspond to the child’s county of residence. Four of the affected counties, Habersham, Henry, Troup, and Whitfield, have operating DV shelters.

tiated child abuse and neglect reports per county-year. The second is the count of reports in which foster care was provided as a post-response service, defined as any out-of-home placement for more than 24 hours, which serves as a measure of severe family disruption. The third outcome is the number of reports in which domestic violence is flagged as present in the home, regardless of whether the child is the direct victim or whether the report is substantiated. This measure reflects children’s potential exposure to domestic violence, which may affect child welfare even when no direct maltreatment is confirmed. All three measures are scaled per 100,000 county population under 18.

Table 1 reports descriptive statistics for the violence and child maltreatment outcomes used in the analysis, all measured at the county-fiscal-year level over the 2013–2019 period. As discussed above, IPV and DV homicide and discharge measures are scaled by total county population, female homicide and assault measures are scaled by female population ages 18–34, and child homicide and NCANDS measures are scaled by county population under 18. Homicides are rare events: the mean county-year IPV homicide count is less than one, whereas hospitalization rates exhibit substantially greater variation across counties.

Table 1: Violence Outcomes Descriptive Statistics

Variable	N	Mean	SD	Min	Max
<i>Panel A: UCR Homicide Reports</i>					
IPV homicides (per 100,000)	308	0.625	1.226	0	9.346
DV homicides (per 100,000)	308	1.017	1.484	0	9.346
Female homicides (per 100,000 females 18–34)	308	2.740	6.776	0	57.400
Child homicides (per 100,000 population under 18)	308	1.462	3.379	0	31.197
<i>Panel B: HCUP Hospital Discharges</i>					
IPV discharges (per 100,000)	308	6.059	6.319	0	32.069
DV discharges (per 100,000)	308	12.860	12.234	0	76.192
Female assault discharges (per 100,000 females 18–34)	308	536.454	257.097	77.027	1402.761
<i>Panel C: NCANDS Child Maltreatment Reports (per 100,000 population under 18)</i>					
Substantiated reports, infants	308	103.741	78.473	0	353.029
Substantiated reports, ages 1–3	308	126.470	108.880	0	518.176
Substantiated reports, ages 4–10	308	265.663	227.729	0	1002.753
Substantiated reports, ages 11–18	308	167.676	139.174	0	709.503
Substantiated reports, all ages	308	663.551	531.173	0	2511.162
DV risk factor	308	225.736	228.304	0	1283.101
Foster care services	308	202.576	161.734	0	741.361

Notes: Observations are at the county-fiscal-year level. The data cover 2013–2019. IPV and DV homicide and discharge counts are scaled per 100,000 county population. Female homicide and assault discharge rates are per 100,000 female population ages 18–34. Child homicide rates and all NCANDS measures are per 100,000 county population under age 18.

A Note on Sample Periods

The data sources used in this paper cover different time periods. The 990 shelter panel covers 2013 to 2023. The UCR homicide, HCUP discharge, and NCANDS data cover 2013 to 2019, with the sample ending before the COVID-19 pandemic to avoid contamination from the sharp rise in homicides and changes in hospital utilization during that period. The CJCC utilization data cover 2017 to 2022 for victim counts and 2018 to 2022 for the service breakdowns, capturing a period of only moderate funding variation. Results from each analysis should therefore be interpreted in light of the funding environment and sample period relevant to that outcome.

3.6 The Georgia Domestic Violence Shelter Landscape

Georgia’s 48 DV shelters operate across 45 of the state’s 159 counties, with DeKalb County the only county served by two shelters. The remaining 114 counties have no shelter. Among the state’s largest metropolitan counties (RUCC codes 1–3), 26 of 74 (35 percent) host at least one shelter, compared with 16 percent of small-town and rural counties (RUCC codes 6–9). Table 2 reports baseline characteristics for shelter and non-shelter counties prior to the 2015 CVF expansion. Counties with shelters are roughly twice as populous as counties without shelters on average and have modestly higher average incomes and lower poverty rates, consistent with shelters being concentrated in more urban areas. Racial composition is nearly identical across the two groups. Baseline female homicide rates and female assault hospitalization rates are somewhat lower in shelter counties than in non-shelter counties, but these differences are not statistically significant. These descriptives are consistent with shelters locating in more populous areas, rather than in areas with higher baseline violence.

Table 2: County Characteristics by Shelter Presence (Pre-2016)

	No Shelter ($N = 114$)		Shelter ($N = 45$)		
	Mean	SD	Mean	SD	Difference
<i>Demographics</i>					
Total population	69,074	176,443	123,397	141,928	54,232** (19,277)
Per capita income (\$)	33,000	7,527	34,484	6,414	1,483* (842)
Median HH income (\$)	42,402	10,787	45,869	12,663	3,466** (1,421)
Poverty rate (%)	22.13	6.89	20.29	6.78	−1.84** (0.82)
Percent female (%)	50.83	2.19	51.19	1.37	0.36* (0.22)
Percent white (%)	68.38	15.60	68.16	19.46	-0.22 (2.13)
<i>Baseline violence rates</i>					
Female homicide (per 100k Female 18-34)	4.39	11.90	3.73	7.59	−0.65 (1.25)
Female assault discharges (per 100k Female 18-34)	565.54	289.37	545.73	247.64	−19.81 (32.44)

Notes: Means are computed over county-year observations from 2013–2015, prior to the 2015 CVF expansion. The difference column reports the shelter-county mean minus the no-shelter-county mean, with heteroskedasticity-robust standard errors from a county-year OLS regression in parentheses. Female homicide and assault discharge rates are per 100,000 female population ages 18–34. Female assault discharge rates are per 100,000 female population ages 18–34. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 1 shows the geographic distribution of government grant funding across Georgia’s DV shelters in 2013 (baseline) and the change in government grants between 2013 and 2018. Funding levels vary substantially across shelters at baseline, and grant trajectories

differ considerably across shelters over the sample period.

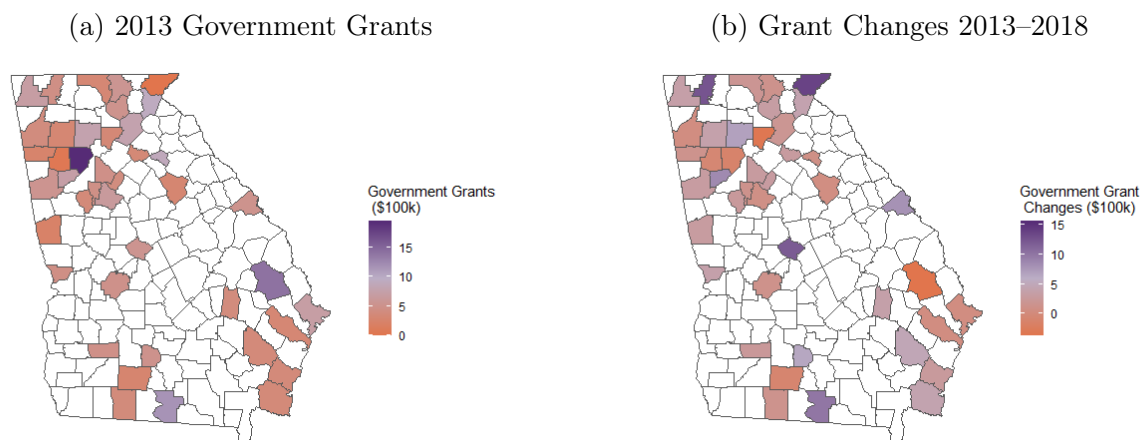


Figure 1: Government Grant Distribution and Changes Across Georgia DV Shelters

Notes: The left panel depicts the total government grants for each DV shelter in Georgia during the 2013 fiscal year. The right panel depicts the change in government grants between 2013 and 2018. Counties in white do not have a DV shelter operating. Houston County (Warner Robins) has a DV shelter operating but is excluded from the financial analysis because the shelter is a small program within a large, multi-service Georgia nonprofit whose financials are not comparable to those of stand-alone DV shelters. Data come from the DV Shelter Database.

Georgia's shelter funding grew substantially over the sample period, as shown in Table 3. Mean total revenue increased 86 percent, from \$974,449 in 2013 to \$1,814,542 in 2023, and government grants grew similarly (up 87 percent, from \$631,170 to \$1,177,207). Private contributions also increased (up 82 percent). Expenditures tracked revenue closely: total expenditure rose 83 percent, and compensation, the largest expenditure category, rose 86 percent. Employee counts grew more modestly (up 36 percent, from 22.5 to 30.6 on average), while reported volunteer counts declined by 49 percent; the volunteer decline should be interpreted cautiously given inconsistent reporting of this field across organizations and years. No shelters entered or exited over the 2013–2023 period, indicating that growth occurred within existing programs rather than through sector expansion.

Table 3: Georgia DV Shelter Financial Characteristics, 2013 and 2023

	2013			2023			
	N	Mean	Median [p25, p75]	N	Mean	Median [p25, p75]	% Change (Mean)
Revenue (\$)							
Total Revenue	47	974,449	732,957 [527,896, 1,089,400]	47	1,814,542	1,389,710 [944,244, 2,119,760]	+86%
Government Grants	45	631,170	472,982 [356,381, 703,990]	47	1,177,207	860,027 [617,613, 1,419,811]	+87%
Private Donations	45	256,566	128,681 [71,366, 242,623]	47	466,576	270,119 [153,778, 544,034]	+82%
Program Service Revenue	46	32,483	0 [0, 5,515]	47	60,143	0 [0, 0]	+85%
Other Revenue	45	75,398	25,968 [1,318, 112,345]	47	110,618	51,974 [13,455, 163,826]	+47%
Expenditure (\$)							
Total Expenditure	47	894,803	705,363 [507,347, 1,021,463]	47	1,634,478	1,253,736 [888,599, 1,954,435]	+83%
Compensation	47	510,133	423,953 [323,359, 566,941]	47	947,329	795,955 [553,326, 1,057,586]	+86%
Fee-for-Service	47	37,855	5,000 [0, 21,654]	47	45,296	15,657 [2,400, 35,828]	+20%
Operations	47	29,035	14,869 [5,445, 27,531]	47	70,482	23,626 [11,714, 81,420]	+143%
Occupancy	47	52,166	44,522 [21,225, 60,302]	47	87,837	70,489 [48,855, 122,305]	+68%
Fundraising	45	35,623	8,067 [0, 19,337]	47	87,946	14,265 [0, 41,439]	+147%
Other Expenditure	43	239,701	163,480 [85,042, 288,413]	47	395,587	400,349 [186,678, 570,191]	+65%
Staffing							
Employees	44	22.5	23.0 [15.5, 27.0]	46	30.6	28.5 [20.0, 37.0]	+36%
Volunteers	31	78.5	69.0 [25.0, 96.0]	26	39.7	27.5 [22.0, 50.0]	−49%

Notes: Each observation is a shelter-year. The analysis sample includes 46 shelters with available Form 990 data for the relevant year; one additional shelter in the 990 database operates as a small program within a large, multi-service Georgia nonprofit and is excluded. Sample sizes vary across variables due to item-level missingness in individual Form 990 entries, so the revenue and expenditure subcategories do not sum exactly to their respective totals in every shelter-year. Percent change is computed as the change in the cross-sectional mean from 2013 to 2023. Program Service Revenue has a median and 75th percentile of zero in both years, indicating that the majority of shelters report no revenue from this source; the small positive mean reflects a subset of shelters with fee-based programming. Volunteer counts should be interpreted with caution: the number of shelters reporting volunteers declined from 31 in 2013 to 26 in 2023, and volunteer reporting on Form 990 is inconsistent across organizations and years.

3.7 Service Provision Across Georgia Shelters, 2018–2022

I use the CJCC service records described in Section 3.2 to characterize which services Georgia shelters provide during 2018–2022, the period for which shelter-by-service data are available. This period follows the 2015 Crime Victims Fund cap increase described in Section 2.2. I characterize service provision for this period only; the patterns below should not be extended to the full 2013–2023 sample used elsewhere in the paper.

For each of 39 individual services,¹⁹ I construct two shelter-level measures: whether a shelter offers the service at any point during 2018–2022, and whether a shelter offers the service in every year it is observed. All 46 shelters with CJCC service records offer the same fourteen services at some point during this period: a staffed crisis line, crisis coordination, safe housing, emergency and non-emergency legal assistance, emergency financial assistance, assistance applying for TANF and other government benefits, individual support, support groups, interagency advocacy, referral services, relocation assistance, transportation assistance, and criminal-justice-system advocacy. Table 4 reports the offering rates underlying this comparison. At the other end of the distribution, four services are offered by fewer than one-fifth of shelters at any point during the sample period: attorney-provided immigration assistance and three categories of forensic medical examination. These services plausibly require partnerships with credentialed providers, such as immigration attorneys or Sexual Assault Nurse Examiners, that most shelters do not maintain in-house.

The remaining services fall between these two groups. Offering rates vary substantially across shelters even within this middle range. Civil legal attorney assistance for family law, for example, is offered by 71.7 percent of shelters at some point during the sample period but by only 6.5 percent of shelters in every year they are observed, a gap of 65 percentage points. Group therapy shows a similar pattern: 58.7 percent of shelters offer it at some point, compared with 4.3 percent that offer it every year.

This gap between ever offering a service and offering it consistently is not limited to less common services. Even among the fourteen near-universal services, the share of shelters offering a service in every observed year is substantially lower than the share offering it in at least one year. All 46 shelters offer Support Group at some point, but only 84.8 percent offer it every year; all 46 offer assistance applying for TANF and other government benefits at some point, but only 63.0 percent offer it every year. Within-shelter service offerings therefore fluctuate from year to year rather than remaining fixed, even for services offered

¹⁹I exclude one additional catch-all category, “Other Service Provided,” because no shelter reports offering it during the sample period.

nearly universally at some point during the sample period.

I also examine the number of victims served annually through each service.²⁰ Even among the fourteen near-universal services, annual victim counts vary widely across shelters. Shelters report serving an average of 7,599 victims annually through Individual Support. The standard deviation, 9,926, exceeds the mean, reflecting large differences in scale across shelters. Table 4 reports mean victims served for the near-universal and least-common service categories; the full 39-service breakdown, including offering rates and victims served for every service, appears in Appendix Table 20.

Table 4: Georgia Shelter Service Provision, 2018–2022

Service	Ever Offered (%)	Always Offered (%)	Mean Victims/Yr
<i>Panel A: Near-Universal Services (offered by all 46 shelters)</i>			
Crisis Line	100.0	95.7	2,017
Individual Support	100.0	95.7	7,599
Referral to Services, Support, Resources	100.0	95.7	1,107
Safe Housing	100.0	91.3	3,684
Transportation Assistance	100.0	93.5	1,297
Support Group	100.0	84.8	617
Non-Emergency Legal Assistance	100.0	73.9	523
Interagency Advocacy	100.0	76.1	823
Emergency Legal Assistance	100.0	73.9	801
Emergency Financial Assistance	100.0	69.6	178
Criminal Justice Support/Advocacy	100.0	69.6	327
Relocation Assistance	100.0	67.4	210
Crisis Coordination	100.0	60.9	2,581
TANF/Government Benefits Assistance	100.0	63.0	110
<i>Panel B: Least-Common Services (offered by fewer than 20% of shelters)</i>			
Performance of Initial Forensic Medical Exam	19.6	8.7	9.4
Performance of Forensic Interview	17.4	8.7	8.8
Performance of Follow-Up Forensic Medical Exam	17.4	2.2	0.7
Immigration Assistance–Attorney	13.0	0.0	6.9

Notes: Sample includes 46 shelters with CJCC service records, 2018–2022 ($N = 230$ shelter-years). “Ever Offered” is the share of shelters offering the service in at least one sample year; “Always Offered” is the share offering it in every year observed. Mean victims/year is the shelter-year average. Data come from CJCC utilization reports. The full 39-service breakdown appears in Appendix Table 20.

²⁰As noted in Section 3.2, victim counts may include duplicates, since an individual receiving multiple services or seeking services from more than one shelter in the same year is counted multiple times.

4 Empirical Strategy

The empirical strategy exploits within-shelter variation in government grant funding over time. Figure 2 shows the distribution of annual government grants across Georgia DV shelters from 2013 to 2023, with one box-and-whisker plot per year. Two features of the funding environment are worth noting. First, the median grant level — the horizontal line within each box — roughly doubles between 2013–2015 and 2016–2023, with the shift visible beginning in 2016 as the additional federal funds were disbursed to shelters. Second, the interquartile range of grant funding increases visibly after 2015, reflecting CJCC allocation decisions across shelters that produced a wider spread in funding levels rather than a uniform increase.

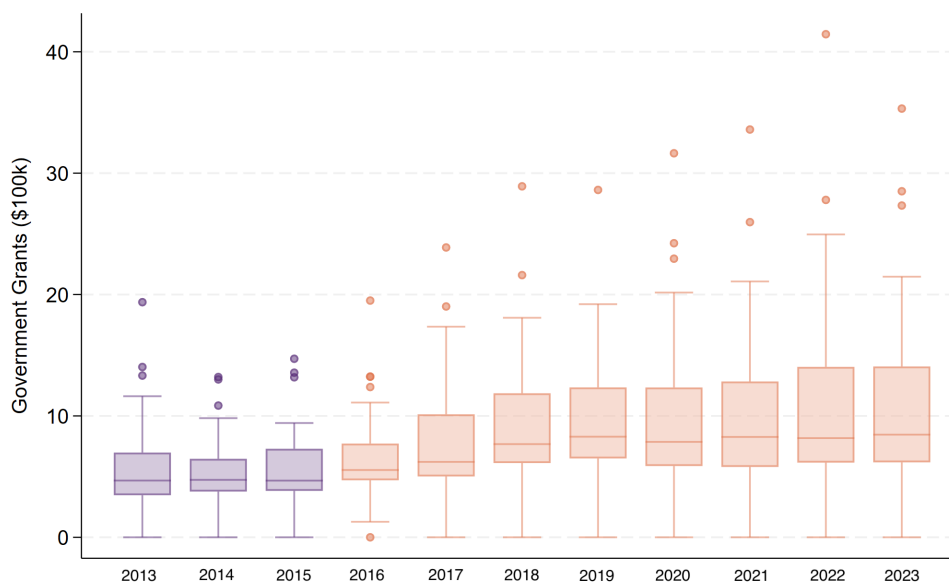


Figure 2: Distribution of Annual Government Grants, 2013–2023

Notes: The figure shows the distribution of annual government grants across Georgia DV shelters from 2013 to 2023, with one box-and-whisker plot per year. Each box spans the interquartile range (25th to 75th percentile) of grant funding in that year, with the horizontal line marking the median; whiskers extend to the most extreme values within 1.5 times the interquartile range, and points beyond this range are plotted individually as outliers. Purple boxes indicate the pre-cap-increase period (2013–2015) and orange boxes indicate the post-cap-increase period (2016–2023). Grants are measured in \$100,000s. Data come from the DV Shelter Database.

4.1 Two-Way Fixed Effects

The primary estimating equation is a two-way fixed effects (TWFE) regression of the form:

$$Y_{it} = \beta_1 \cdot (\text{Grants}/\$100\text{k})_{it} + \beta_2 \cdot (\text{Grants}/\$100\text{k})_{i,t-1} + \alpha_i + \gamma_t + \varepsilon_{it}, \quad (1)$$

where Y_{it} is the outcome of interest for shelter or county i in fiscal year t . Shelter-level outcomes come from the DV shelter database and utilization data while county-level outcomes are the DV and child maltreatment measures. The term α_i denotes shelter or county fixed effects that absorb all time-invariant characteristics such as mission focus or geographic service area, and γ_t denotes year fixed effects that account for aggregate shocks affecting all shelters or counties in a given year. Standard errors are clustered at the shelter or county level to account for serial correlation within units over time.

I include the one-period lag of grant funding, $\text{Grants}_{i,t-1}$, because shelters can carry unspent grant funds into the following fiscal year. Without the lag, current expenditures financed by prior-year grants would be misattributed to current-year funding, causing β_1 to overstate the within-year spending response. Including the lag isolates the contemporaneous effect from this carryover. I interpret β_1 , the coefficient on current-year grants, as the primary coefficient of interest throughout, with β_2 included as a control.

For the county-level specifications, I aggregate shelter grant funding to the county level. In Georgia, DeKalb County is the only county with multiple shelters; for this county I sum government grants across shelters to measure total funding allocated to the county.

Equation (1) is estimated separately for three categories of outcomes: (i) revenue and expenditure; (ii) organizational; and (iii) county-level violence and child maltreatment.

4.2 Identification

The setting differs from much of the recent difference-in-differences literature in an important respect: treatment is continuous. The forbidden-comparisons problem identified in Callaway

and Sant’Anna (2021) and related work arises because already-treated units serve as implicit controls for later-treated units under staggered binary treatment designs, generating negatively weighted comparisons when treatment effects are heterogeneous over time or treatment cohort.

A growing econometric literature has developed estimators for continuous or staggered treatments, including Callaway et al. (2024) and de Chaisemartin et al. (2024). However, these estimators are not well suited to this setting: Callaway et al. (2024) requires never-treated or not-yet-treated units, which do not exist here since every shelter receives funding every year, while de Chaisemartin et al. (2024) accommodates continuous treatment but not staggered timing, and relies on a no-carryover assumption that is difficult to justify given persistent, cumulative funding. I therefore use TWFE with a lagged control as the baseline specification.

Recovering a causal interpretation of β_1 in equation (1) requires four conditions to hold. I describe each condition below, then present the evidence I use to assess it.

4.2.1 Assumptions

Parallel trends. The core identifying assumption is that shelters with different levels of government grant funding would have evolved similarly in the absence of treatment. With continuous treatment, this is a stronger requirement than the standard binary-treatment version of the parallel trends assumption: it requires that units receiving different funding doses would have followed parallel paths absent the variation in grants. This assumption would be violated if the CJCC directs more funding toward shelters in counties with higher DV rates, worsening trends, or greater organizational need, since then within-shelter variation in grants would reflect these underlying conditions rather than an exogenous funding change. The CJCC’s allocation criteria are not publicly documented, which makes this assumption difficult to assess directly.

Linearity. The TWFE specification assumes a linear dose-response relationship be-

tween grant funding and each outcome. This assumption would be violated if the marginal effect of funding varies with the level of funding. For example, funding may exhibit diminishing returns once a shelter’s core staffing and capacity needs are met, or increasing returns if small grants are insufficient to finance meaningful operational changes.

Homogeneous treatment effects. The specification also assumes that the effect of a given funding increase is constant across shelters and over time. This assumption would fail if, for example, larger or more established shelters are better positioned to convert additional funding into program expansion than smaller shelters.

I relax the linearity and homogeneous treatment effects assumptions by estimating the model separately by baseline shelter size, defined by terciles of 2013 total revenue. This allows the dose-response relationship and treatment effects to vary across the shelter size distribution without imposing a specific functional form on those differences. This approach also has limitations. Because each tercile contains relatively few shelters, the resulting estimates are imprecise, and the terciles provide only a coarse approximation to what is likely a continuous relationship between shelter size and treatment effects. Accordingly, I interpret the heterogeneity analysis as exploratory and treat any differences across terciles cautiously. The aggregate specification nevertheless continues to rely on the assumptions of linearity and homogeneous treatment effects.

Limited carryover. The specification assumes that including a single lag of grant funding is sufficient to capture carryover in unspent funds. This is a weaker requirement than assuming no carryover at all, since the lagged term absorbs one year of delayed spending, but it would be violated if shelters carry funds forward more than one year. If carryover persists beyond a single lag, β_1 would continue to partially reflect delayed spending from grants received more than one year prior.

4.2.2 Evidence

I provide two pieces of evidence that support the parallel trends assumption following Alexander and Schnell (2024). First, I estimate balancing regressions using equation (1) with county-level determinants of DV and lagged DV incidents as outcomes. The results, shown in Table 5, show no evidence that government funds are targeted toward shelters in counties with more at-risk populations or higher prior DV rates. These results suggest that CJCC allocation decisions are not systematically related to observed measures of county-level need, though a balancing test of this kind cannot rule out targeting based on unobserved characteristics correlated with funding but not captured by the observed county controls.

Second, I estimate an event study specification. Let $\Delta\text{Grants}_i = \text{Grants}_{i,2016} - \text{Grants}_{i,2015}$ denote the change in government grant funding for shelter i between 2015 and 2016. I use 2016 as a quasi-event year because it is the first year in which the sustained VOCA funding shift appears in shelter-level grant data. This choice minimizes the extent to which pre-period coefficients capture the effects of the funding increase. However, not all shelters experience their largest funding change in 2016; given the idiosyncratic nature of CJCC allocation decisions, shelters receive funding increases at different points across the post-2015 period. The event study therefore captures the dynamic response to the earliest funding change after the VOCA cap increase rather than the full trajectory of the sustained funding environment. I estimate:

$$Y_{it} = \sum_t \beta_t \cdot \Delta\text{Grants}_i \cdot \rho_t + \alpha_i + \gamma_t + \epsilon_{it} \quad (2)$$

where ρ_t are year indicators and β_t captures the differential trend in outcomes for shelters that experienced larger initial funding changes. The pre-period coefficients ($t < 2016$) provide evidence on parallel trends: if shelters that received larger initial funding increases were already on different trajectories before the sustained funding shift, the identifying assumption would be violated. The post-period coefficients trace the dynamic response to the funding

Table 5: Balancing Regressions

	β	Mean
Total Births	-9.7666*** (2.6830)	2068.15 (2775.62)
Per Capita Income	74.9482 (70.3913)	37405.44 (7789.87)
Median HH Income	-87.9181 (101.9412)	49788.45 (14232.65)
Pct. in Poverty	-0.0192 (0.0598)	18.36 (6.49)
Unemployment Rate	-0.0008 (0.0207)	5.80 (1.93)
Pct. Female	0.0073 (0.0059)	51.29 (1.42)
Pct. White	0.0280 (0.0536)	66.20 (20.51)
Pct. Black	-0.0056 (0.0483)	28.82 (19.86)
Pct. Native American	-0.0024 (0.0025)	0.56 (0.23)
Pct. Asian	-0.0163 (0.0256)	2.35 (2.20)
Pct. Two or More Races	-0.0034 (0.0021)	1.93 (0.61)
Lagged IPV Homicide	0.0095 (0.0138)	0.76 (1.20)
Lagged DV Homicide	0.0226 (0.0166)	1.32 (1.51)
Lagged Female Homicide	0.0531 (0.0904)	3.13 (5.79)
Lagged IPV Discharges	0.0738 (0.1789)	5.81 (6.17)
Lagged DV Discharges	0.0779 (0.2739)	11.91 (11.88)
Lagged Assault Discharges	1.1144 (3.1787)	428.86 (203.94)

Notes: The table presents the associations between a \$100,000 increase in annual government grant funding for shelters and possible confounders. The coefficients are estimated using equation (1) with county and year fixed effects. County demographics come from the Census and Bureau of Labour Statistics, homicides are measured using UCR-SH, and discharges from the GA HCUP databases. DV incidents are lagged by a year to estimate the relationship between current funding and the previous year's violence.

change.



Figure 3: Event Study Estimates of 2016 Government Grant Change

Notes: The figures plot the coefficients and 95 percent confidence intervals on the interaction between the initial shelter grant change and year indicators from equation (2). Financial outcomes are drawn from the 990 DV shelter database and cover 2013–2023. Total revenue and expenditure are measured in \$100,000. Violence outcomes are drawn from GA HCUP discharge records and UCR-SH reports and cover 2013–2019.

The pre-period coefficients for the financial outcomes are small and statistically indistinguishable from zero, consistent with the parallel trends assumption for the revenue and expenditure specifications. The pre-period event-study coefficients for the violence outcomes are imprecisely estimated likely because county-level violence outcomes are rare. Consequently, the event study has limited power to detect pre-treatment differences, so the absence of statistically significant pre-trends provides weaker evidence in support of the identifying assumptions for these particular outcomes. The balance table evidence, showing no correla-

tion between current funding and lagged violence, provides complementary support for the identifying assumption in the violence specifications.

Finally, I decompose variation in government grant funding, G_{it} , into components explained by permanent differences across shelters, by common year effects, and by both. Table 6 reports the results. Shelter fixed effects alone account for 68.7 percent of the variation in grant funding, while year fixed effects alone account for 13.5 percent. Grant funding is therefore best characterized as varying primarily across shelters rather than as a common, statewide expansion that affected all shelters similarly in a given year.

Table 6: Variance Decomposition of Grant Funding

Specification	R^2	N
(1) No fixed effects	0.000	494
(2) Shelter fixed effects	0.687	494
(3) Year fixed effects	0.135	494
(4) Shelter and year fixed effects	0.819	494

Notes: Standard errors clustered by shelter.

Additional evidence on the CJCC allocation process including tests of whether lagged shelter finances, county violence, or county demographics predict grant levels or grant changes, and whether changes in violence or child maltreatment outcomes predict future funding is presented in Appendix C.

Taken together, this evidence mitigates the concern that CJCC allocation decisions are systematically related to time-varying unobserved shelter or county characteristics, supporting a causal interpretation of β_1 .

5 Results

This section presents the main results. I first examine the effects of government grant funding on shelter finances and staffing. I then turn to community-level outcomes, examining effects on violence and child maltreatment.

5.1 Effects on Shelter Finances

The estimates in Table 7 suggest that a \$100,000 increase in government grants raises a shelter’s annual revenue by \$61,900 (61.9 cents per grant dollar). The results suggest revenue crowd-out is driven by private donations with a decline of \$38,900 with a \$100,00 increase in government grants. This level of crowd-out contrasts with the nonprofit literature, in which government grants displace an estimated 70–75 cents of private giving per dollar (Andreoni & Payne, 2003, 2011). The point estimates for program and other revenue are not statistically distinguishable from zero though the confidence intervals can not rule out up to a 16% increase in program revenue. The crowd-out pattern is also shown in the estimates for revenue shares with a 3.1% increase in the share of revenue grants and a 2.7% decrease in the share of private donations. Contemporaneous annual grants increase total revenue with evidence of limited private donation crowd-out.

Table 7: Effects of Government Grants on DV Shelter Revenue

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Revenue Level (\$100k)				Share of Total Revenue			
	Total	Private	Program	Other	Grants	Private	Program	Other
	Revenue	Donations	Revenue	Revenue		Donations	Revenue	Revenue
Grants (\$100k)	0.619***	-0.389**	0.021	-0.013	0.031***	-0.027***	-0.000	-0.003
	(0.147)	(0.161)	(0.022)	(0.030)	(0.007)	(0.007)	(0.001)	(0.002)
Grants (lagged)	0.378**	0.397**	0.008	-0.028	-0.010***	0.014***	0.001	-0.004
	(0.158)	(0.170)	(0.012)	(0.036)	(0.003)	(0.005)	(0.001)	(0.004)
Sample Mean	12.888	3.046	0.273	0.857	0.680	0.226	0.022	0.072
Observations	442	442	442	442	442	442	442	442

Notes: The table shows the estimated effects of a \$100,000 increase in government grants, while controlling for the prior year grant amount. Observations are at the shelter-fiscal-year level from 2013–2023. All regressions include shelter and year fixed effects. Private donations comprise fundraising event revenue and direct contributions (IRS Form 990 Part VIII lines 1c and 1f). Program revenue is reported on Part I of Form 990. Other revenue includes federated campaigns, membership dues, grants from other organizations, and miscellaneous receipts; this category can be negative for shelter-years with investment losses. Data come from the DV Shelter Database. Standard errors in parentheses are clustered at the shelter level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

I then examine the effect of increased government grants on organization expenditure. I present the results for observed expenditure categories in Table 8. A \$100,000 increase in grants leads to a \$57,300 increase in total expenditure. This estimate reflects an expenditure pass-through of 57 cents per grant dollar, compared to the 62 cent revenue pass-through. This small gap could reflect organizations saving a portion of grant funds rather than spending them the same year. Organizations assign 80 percent of their increased expenditure to their primary program, the functional category shelters use to classify their predominant service offerings. To examine spending patterns in more detail, I categorize total expenditure into six mutually exclusive bins and report the estimated effects in columns (3)–(8) of Table 8. I find that a \$100,000 increase in government grants raises employee compensation by \$31,600, other expenditure by \$13,500,²¹ and, marginally, fee-for-service contract work by \$4,700 ($p = 0.098$). Organizations direct 55 percent of their increased expenditure toward staff compensation, while occupancy expenditure shows no significant effect. Whether this compensation increase reflects expanded staffing capacity is examined directly in Table 9.

²¹Other expenditure includes any spending not captured in the five named categories; because this category spans a wide range of spending types, I do not draw a strong conclusion from this result.

Table 8: Effects of Government Grants on DV Shelter Expenditure

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Primary	Compensation	Contract	Operations	Occupancy	Fundraising	Other
	Expenditure	Program		Work			Expenditure	
	(\$100k)							
Grants (\$100k)	0.573***	0.447***	0.316***	0.047*	0.034	-0.009	0.051	0.135**
	(0.069)	(0.063)	(0.050)	(0.028)	(0.024)	(0.021)	(0.040)	(0.059)
Grants (lagged)	0.252***	0.266***	0.116***	0.020	-0.001	0.040*	0.013	0.064
	(0.051)	(0.038)	(0.039)	(0.043)	(0.013)	(0.020)	(0.010)	(0.046)
Sample Mean	11.989	9.081	7.008	0.513	0.469	0.661	0.519	2.820
Observations	442	442	442	442	442	442	442	442

Notes: The table shows the estimated effects of a \$100,000 increase in government grants, while controlling for the prior year grant amount. Observations are at the shelter-fiscal-year level from 2013–2023. All regressions include shelter and year fixed effects. Compensation includes all W-2 employee compensation — officer and employee salaries, benefits, and payroll taxes (Form 990 Part IX lines 5–10) — and excludes contract workers. Contract work comprises all fee-for-service expenditure lines. Operations includes office, information technology, advertising, travel, and conference expenditures. Other is a residual category capturing supplies, maintenance, bank fees, and miscellaneous items; the six mutually exclusive categories sum to total expenditure by construction. Primary program expenditure reflects shelters’ own functional classification of spending directed toward service delivery and is not mutually exclusive with columns 3–8. Data come from the DV Shelter Database. Standard errors in parentheses are clustered at the shelter level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5.2 Effects on Staffing

Organization staff can play a pivotal role in providing support for victims of domestic violence and can determine what services are offered. Based on the compensation finding, I next examine other characteristics of staffing and present the results in Table 9. Columns (1) and (2) redisplay the estimates from Table 8. I next report two measures of staff composition in columns (3) and (4): total employee count and total volunteer count. A \$100,000 increase in grants raises employee count by 0.411 employees in the same year, controlling for the prior year grant amount, relative to a sample mean of 27.24 employees. As a complementary check, I construct a measure of compensation per employee, defined as total compensation divided by employee count, and estimate its relationship with grant funding. A \$100,000 increase in

grants raises compensation per employee by \$701 in the same year, relative to a sample mean of \$25,423. This result is consistent with grants raising average pay per employee in addition to headcount, though the estimate is only marginally significant. Because compensation per employee is not observed at the individual level, I interpret this result as suggestive rather than definitive. The estimated contemporaneous effect of grants on volunteer count is negative but not statistically significant. The volunteer count regression is estimated on a much smaller sample than the other columns, because fewer shelters report volunteer counts consistently on Form 990. Taken together, employee count rises with grant funding, and compensation per employee shows a similar, though noisier, pattern; volunteer count shows no comparable relationship. These results are consistent with a modest expansion in paid staffing capacity, though not all of the underlying estimates are precisely estimated.

Table 9: Effects of Government Grants on DV Shelter Staffing

	(1)	(2)	(3)	(4)	(5)
	Compensation	Contract	Employee	Volunteer	Compensation per
	(\$100k)	Work (\$100k)	Count	Count	Employee (\$100k)
Grants (\$100k)	0.316*** (0.050)	0.047* (0.028)	0.411* (0.207)	-1.319 (1.581)	0.007** (0.003)
Grants (lagged)	0.116*** (0.039)	0.020 (0.043)	0.503** (0.211)	-2.129* (1.082)	-0.004 (0.003)
Sample Mean	7.008	0.513	27.24	53.98	0.25
Observations	442	442	370	150	340

Notes: The table shows the estimated effects of a \$100,000 increase in government grants, while controlling for the prior year grant amount. Observations are at the shelter-fiscal-year level from 2013–2023. All regressions include shelter and year fixed effects. Compensation includes all W-2 employee compensation — officer and employee salaries, benefits, and payroll taxes (Form 990 Part IX lines 5–10) — and excludes contract workers. Contract work comprises all fee-for-service expenditure lines and captures payments to independent contractors. Compensation per employee is constructed as total compensation divided by employee count and is not observed at the individual level. Data come from the DV Shelter Database. Standard errors in parentheses are clustered at the shelter level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5.3 Effects on Violence Outcomes

Section 2 outlined three channels through which shelter funding could affect community-level violence: improving service quality for existing clients, expanding capacity to reach additional victims, or increasing outreach without expanding capacity. The staffing results in Section 5.2 indicate that shelters do expand capacity in response to funding, with employee counts rising alongside compensation. Whether this expanded capacity produces a detectable effect on community-level violence, however, depends on which of the three channels dominates. These channels apply not only to violence against adult victims but to children’s exposure to violence in the household, since expanded shelter capacity and outreach could affect both populations through the same mechanisms. I study county-year-level counts of DV-related hospital discharges, homicides, and child maltreatment reports, presenting the results in Tables 10, 11, and 12 respectively. Because each table tests multiple outcomes and both a contemporaneous and lagged grant coefficient for each, I apply a Benjamini-Hochberg correction within each data source to assess which results are robust to multiple testing.²²

Table 10 reports the estimated effects of government grants on hospital discharge rates. The contemporaneous coefficients for IPV, DV, and female assault discharges are not statistically distinguishable from zero. For IPV and DV discharges, the confidence intervals are wide relative to the sample mean, ruling out effects larger than roughly 5 to 8 percent; these estimates are therefore uninformative about moderate-sized effects rather than evidence of a precise null. For female assault discharges, the confidence interval is narrower, ruling out contemporaneous effects larger than about 1.5 percent of the sample mean.

The lagged coefficient for female assault discharges is positive and small in magnitude (an increase of roughly 0.5 to 3.2 percent of the sample mean). This is the only violence or child maltreatment result in this section that remains significant after the Benjamini-

²²Corrections are computed separately by data source: 8 tests for UCR homicide measures, 6 tests for HCUP discharge measures, and 14 tests for NCANDS child maltreatment measures, each counting the contemporaneous and lagged coefficient for every outcome shown in the corresponding table. I use $q = 0.10$ to match the significance threshold used throughout the paper.

Hochberg correction. The lagged coefficients for IPV and DV discharges remain statistically indistinguishable from zero.

Table 10: Effects of Government Grants on Hospital Discharges

	(1)	(2)	(3)
	IPV Discharge	DV Discharge	Assault Discharge
	Rate per 100k	Rate per 100k	Rate per 100k
	Total County Population		Female Pop. 18-34
Grants (\$100k)	0.004	-0.058	-2.563
	(0.154)	(0.200)	(2.775)
Grants (lagged)	-0.105	0.012	9.799**
	(0.177)	(0.298)	(3.655)
Sample Mean	5.956	11.979	535.061
Observations	264	264	264

Notes: The table shows the estimated effects of a \$100,000 increase in government grants, while controlling for the prior year grant amount. The dependent variable in each column is a discharge rate expressed per 100,000 population. Corresponding Poisson estimates with a population exposure offset appear in Appendix Table 27. Column (1) identifies discharges with a cause-of-injury code for abuse with an identified perpetrator. Column (2) identifies any domestic violence discharge. Column (3) identifies assault discharges for women ages 18–34. Discharges are aggregated across emergency departments and inpatient facilities in Georgia. All regressions include county and year fixed effects. Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11 reports the estimated effects of government grants on county-level homicide rates. The IPV homicide coefficients are opposite in sign but similar in magnitude: a decline of 11 percent for the contemporaneous coefficient (95 percent CI: 0.6 to 22 percent), and an increase of similar size for the lagged coefficient (95 percent CI: 0.5 to 33 percent). Both coefficients are significant at the 5 percent level, but neither survives the Benjamini-Hochberg correction applied across the eight UCR homicide tests. I find no statistically significant relationship for the remaining homicide measures — DV homicides, female homicides, and child homicides — with either contemporaneous or lagged grants, before or after correction.

Even setting aside the multiple-testing correction, the opposing signs on the IPV homicide coefficients and the width of their confidence intervals make it difficult to draw a

stable conclusion from this result. I do not find a similar pattern in the DV, female, or child homicide measures, nor in the hospital discharge or child maltreatment outcomes.

Table 11: Effects of Government Grants on DV Homicides

	(1)	(2)	(3)	(4)
	IPV Homicide	DV Homicide	Female Homicide	Child Homicide
	Total County Population		Female Pop. 18–34	Pop. Under 18
Grants (\$100k)	-0.064** (0.030)	-0.007 (0.068)	-0.247 (0.258)	-0.036 (0.061)
Grants (lagged)	0.096** (0.046)	0.054 (0.064)	-0.038 (0.206)	0.010 (0.089)
Sample Mean	0.566	0.982	2.541	1.417
Observations	264	264	264	264

Notes: The table shows the estimated effects of a \$100,000 increase in government grants, while controlling for the prior year grant amount. The dependent variable in each column is a homicide rate expressed per 100,000 population. Corresponding Poisson estimates with a population exposure offset appear in Appendix Table 28. IPV, DV, and female homicide outcomes come from UCR-SH data identified by perpetrator codes and victim sex. Child homicide counts any homicide with a victim under age 18, regardless of perpetrator relationship. All regressions include county and year fixed effects. Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 12 reports the estimated effects of government grants on substantiated child maltreatment reports, foster care placements, and reports flagging domestic violence in the household. Six of the seven contemporaneous and lagged coefficient pairs are not statistically distinguishable from zero. The one exception is the contemporaneous coefficient for substantiated reports among children ages 11–18, which is negative at the 10 percent level (a decline of 3.419 per 100,000, relative to a sample mean of 166.506) but does not survive the Benjamini-Hochberg correction applied across the fourteen NCANDS tests. I do not find a similar pattern in the corresponding lagged coefficient or the all-ages aggregate measure. As with the IPV homicide result, I do not treat this single, uncorroborated coefficient as evidence of an effect.

Table 12: Effects of Government Grants on Child Maltreatment Reports

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	All Ages	Infant	Ages 1–3	Ages 4–10	Ages 11–18	DV Risk	Foster
	Substantiated Reports					Factor	Services
	Rate per 100k Population Under 18						
Grants (\$100k)	-8.834	-0.639	-1.704	-3.072	-3.419*	-2.565	0.219
	(7.461)	(1.224)	(1.726)	(3.599)	(1.908)	(2.959)	(3.122)
Grants (lagged)	10.255	0.652	2.275	4.949	2.380	-0.285	4.534
	(10.431)	(1.279)	(2.124)	(4.769)	(2.876)	(2.977)	(3.982)
Sample Mean	655.917	104.570	123.325	261.516	166.506	235.576	199.883
Observations	264	264	264	264	264	224	264

Notes: The table shows the estimated effects of a \$100,000 increase in government grants, while controlling for the prior year grant amount. The dependent variable in each column is a substantiated-report rate expressed per 100,000 county population under 18, except column (6), which uses the count of reports flagging domestic violence as present in the household regardless of substantiation. Corresponding Poisson estimates with a population exposure offset appear in Appendix Table 29. All regressions include county and year fixed effects. Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Of the violence and child maltreatment outcomes examined in this section, only the lagged effect on female assault discharges is significant after correcting for multiple testing. The IPV homicide result does not survive correction and displays opposing contemporaneous and lagged signs. Taken together, these results do not point clearly toward any one of the three channels described in Section 2: the surviving evidence is limited to a single, small-magnitude result, and the remaining estimates are too imprecise or too inconsistent to distinguish improved service quality, expanded capacity, or outreach as the dominant mechanism.

5.4 Heterogeneity by Shelter Size

The aggregate results mask potential heterogeneity across shelters of different sizes. Because smaller shelters operate on narrower budgets, equivalent dollar increases in grants represent a larger change relative to baseline revenue. Therefore, I estimate the model separately by baseline shelter size, defined by 2013 revenue terciles. Table 13 reports the resulting cutoffs and summary statistics for each tercile, and Figure 4 shows that funding growth was not uniform across size groups.

Table 13: Size Tercile Cutoffs

Tercile	<i>N</i>	Total Revenue 2013			
		Mean	Std. Dev.	Min	Max
Small	16	\$440,218	\$112,917	\$179,782	\$564,403
Medium	15	\$724,836	\$109,233	\$572,625	\$935,437
Large	15	\$1,474,837	\$652,383	\$992,044	\$2,740,121

Notes: Terciles are defined based on each shelter’s total revenue in 2013.

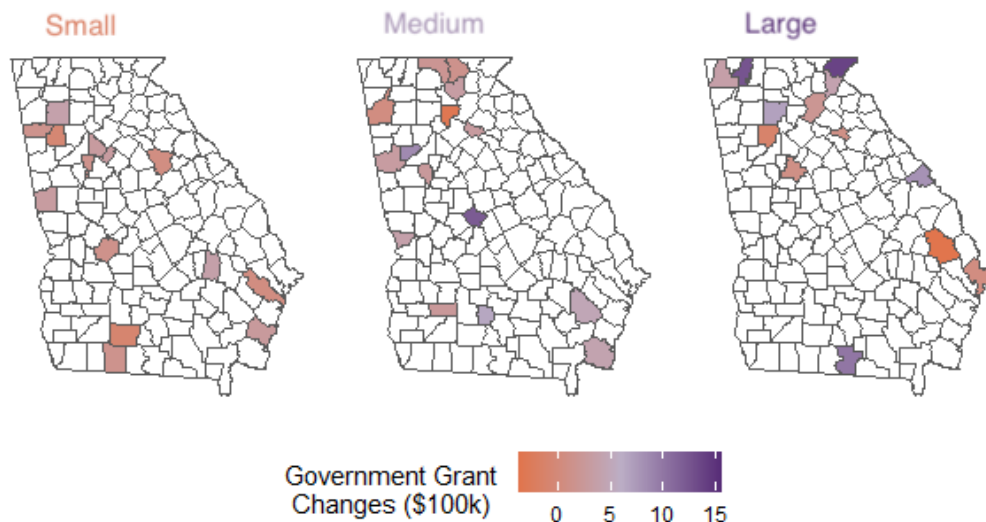


Figure 4: Grant Changes 2013–2018 Split by 2013 Revenue Tercile

Notes: This figure depicts the change in the total government grants for each DV shelter in Georgia between the 2013 and 2018 fiscal year. White counties do not have a DV shelter operating. Shelter sizes are defined by 2013 revenue terciles. Data come from the DV Shelter Database.

Estimating equation 1 separately for each tercile allows the dose-response relationship and year effects to vary freely across size groups. This flexibility also makes it easy to overstate a size-based story by comparing significance across separate regressions rather than testing whether the underlying coefficients significantly differ from one another. I address this directly with a pooled specification that interacts tercile indicators with the contemporaneous and lagged grant terms, testing equality of the tercile-specific coefficients for every outcome examined in this section. Because this pooled test requires the stronger assumption of common time trends across shelter sizes, I treat it as a complement to, rather than a replacement for, the separate-regression estimates that follow.

Table 14 reports these joint tests for every outcome examined in this section, with Benjamini-Hochberg corrections applied separately within each outcome group — Revenue, Expenditure, Staffing, hospital discharges, homicide measures, and child maltreatment measures — using $q = 0.10$ to match the significance threshold used throughout the paper.

Table 14: Testing Equality of Grant Effects Across Shelter Size Terciles

Outcome	p (joint test)	p (BH-adjusted)	
<i>Panel A: Revenue</i>			
Total Revenue	0.006	0.044	**
Private Donations	0.011	0.046	**
Program Revenue	0.334	0.564	
Other Revenue	0.516	0.590	
Grant Share	0.352	0.564	
Private Donation Share	0.340	0.564	
Program Revenue Share	0.431	0.574	
Other Revenue Share	0.800	0.800	
<i>Panel B: Expenditure</i>			
Total Expenditure	0.000	0.003	***
Primary Program Expenditure	0.003	0.011	**
Compensation	0.015	0.041	**
Operations	0.044	0.088	*
Contract Work	0.099	0.159	
Other Expenditure	0.158	0.211	
Fundraising Expenditure	0.316	0.361	
Occupancy	0.440	0.440	
<i>Panel C: Staffing</i>			
Volunteer Count	0.012	0.024	**
Employee Count	0.588	0.588	
<i>Panel D: HCUP Hospital Discharges</i>			
IPV Discharge Rate	0.014	0.041	**
DV Discharge Rate	0.033	0.049	**
Female Assault Discharge Rate	0.586	0.586	
<i>Panel E: UCR Homicide Measures</i>			
Child Homicide Rate	0.058	0.202	
Female Homicide Rate	0.101	0.202	
IPV Homicide Rate	0.507	0.564	
DV Homicide Rate	0.564	0.564	
<i>Panel F: NCANDS Child Maltreatment Measures</i>			
NCANDS DV Risk Factor	0.020	0.137	
Age 11–18 Substantiated Reports	0.075	0.184	
Infant Substantiated Reports	0.079	0.184	
Age 1–3 Substantiated Reports	0.392	0.685	
All Ages Substantiated Reports	0.504	0.705	
Foster Care Services	0.840	0.883	
Age 4–10 Substantiated Reports	0.883	0.883	

Notes: Each row tests whether the effect of a \$100,000 increase in government grants differs across small, medium, and large shelter terciles, using a pooled specification that interacts tercile indicators with the contemporaneous and lagged grant terms, with shelter or county and year fixed effects and standard errors clustered at the shelter or county level. Benjamini-Hochberg corrections are applied separately within each panel, using $q = 0.10$.

Nine outcomes reject equality across shelter size after correction, concentrated in shelter finances, staffing, and hospital discharge rates. I find no significant size heterogeneity for any violence or child maltreatment outcome in the UCR homicide or NCANDS panels. Table 15 reports the separate-regression estimates, including the lagged grant coefficient, for each of these nine outcomes; full tercile-specific results for every outcome examined in this section are reported in Appendix E.

Revenue

In Table 15, a \$100,000 increase in grants raises total revenue by \$59,000 for small shelters, \$108,100 for medium shelters, and \$79,800 for large shelters. A pooled specification with common year effects rejects equal revenue effects across size groups after correction ($p = 0.044$): medium shelters differ significantly from both small ($p = 0.041$) and large ($p = 0.003$) shelters, while small and large shelters are statistically indistinguishable from each other ($p = 0.336$). The revenue heterogeneity result is therefore specifically that medium shelters see a larger revenue increase per grant dollar than either smaller or larger shelters, not a monotonic gradient by size.

One contributor to medium shelters' larger revenue response may be the absence of crowd-out at that size. Table 15 shows a significant decline in private donations for small shelters (\$31,700 per \$100,000 in grants), while medium shelters have a small, precisely estimated coefficient close to zero, indicating little to no crowd-out. The joint test also rejects equal effects by size for private donations ($p = 0.046$), and this pattern is consistent with small shelters driving the result: small and medium shelters differ significantly in the pairwise tests ($p = 0.053$), while large shelters' comparatively imprecise estimate makes the medium-versus-large pairwise difference ($p = 0.008$) difficult to interpret as a distinct crowd-out pattern among large shelters.

Table 15: Effects of Government Grants by Shelter Size: Outcomes with Significant Heterogeneity

	Total Revenue (\$100k)	Private Donations (\$100k)	Total Expenditure (\$100k)	Primary Program (\$100k)	Compen- sation (\$100k)	Opera- tions (\$100k)	Volunteer Count	IPV Discharge Rate	DV Discharge Rate
<i>Panel A: Small Shelters</i>									
Grants (\$100k)	0.590*** (0.214)	-0.317** (0.160)	0.404** (0.206)	0.398*** (0.112)	0.245* (0.133)	0.009 (0.007)	-1.628 (1.476)	-0.490* (0.237)	-0.577 (0.394)
Grants (lagged)	-0.061 (0.055)	0.021 (0.074)	-0.020 (0.049)	0.082*** (0.029)	-0.014 (0.025)	0.001 (0.007)	0.488 (0.734)	0.264 (0.357)	0.471 (0.599)
Sample Mean	6.890	1.333	6.459	4.860	3.994	0.220	31.57	5.751	11.974
N	176	168	176	168	176	176	106	90	90
<i>Panel B: Medium Shelters</i>									
Grants (\$100k)	1.081*** (0.053)	0.039 (0.066)	0.959*** (0.045)	0.825*** (0.082)	0.524*** (0.069)	0.086*** (0.020)	-0.307 (3.966)	-0.309** (0.133)	-0.415 (0.550)
Grants (lagged)	0.005 (0.027)	0.032 (0.024)	0.032 (0.024)	-0.014 (0.035)	0.072** (0.031)	0.003 (0.009)	-0.272 (1.242)	0.088 (0.309)	0.366 (0.754)
Sample Mean	10.777	2.283	10.132	7.861	6.130	0.390	65.00	6.395	13.520
N	165	161	165	161	165	165	104	90	90
<i>Panel C: Large Shelters</i>									
Grants (\$100k)	0.798*** (0.114)	-0.193 (0.124)	0.696*** (0.119)	0.562*** (0.110)	0.335*** (0.058)	0.016 (0.020)	-6.919* (3.595)	0.435* (0.228)	0.390 (0.316)
Grants (lagged)	-0.084 (0.086)	-0.070 (0.077)	-0.053 (0.039)	-0.003 (0.035)	-0.063** (0.032)	-0.002 (0.005)	-0.061 (1.348)	-0.381* (0.212)	-0.295 (0.360)
Sample Mean	19.792	5.232	18.215	13.653	10.185	0.719	110.28	5.853	12.735
N	165	165	165	165	165	165	92	84	84

Notes: The table reports the estimated effects of a \$100,000 increase in government grants, adding a one-year lagged grant term to the specification, estimated separately by baseline shelter size tercile (2013 total revenue), for the nine outcomes with statistically significant size heterogeneity after Benjamini-Hochberg correction within outcome group (Table 14). Financial and staffing outcomes are estimated at the shelter level with county and year fixed effects, standard errors clustered at the shelter level; IPV and DV Discharge Rate are estimated at the county level with county and year fixed effects, standard errors clustered at the county level. Full tercile-specific results for every outcome examined in this section, including outcomes without significant heterogeneity, are reported in Appendix E. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Expenditure

A \$100,000 increase in grants raises total expenditure by \$40,400 for small shelters, \$95,900 for medium shelters, and \$69,600 for large shelters. The joint test rejects equal effects by size ($p = 0.003$). This pairwise pattern is a small-versus-larger-shelters split, rather than the medium-outlier pattern found for revenue: small shelters differ significantly from both medium ($p = 0.001$) and large ($p < 0.001$) shelters, while medium and large shelters are statistically indistinguishable from each other ($p = 0.883$).

The same split holds for primary program expenditure and compensation, the two largest components of the total effect. For primary program expenditure, small shelters differ significantly from both medium ($p = 0.001$) and large ($p = 0.007$) shelters, while medium and large shelters do not differ ($p = 0.606$). For compensation, the pattern is nearly identical: small shelters differ from both medium ($p = 0.024$) and large ($p = 0.010$) shelters, while medium and large again do not differ ($p = 0.647$). Across all three outcomes, small shelters have a smaller expenditure response per grant dollar than either medium or large shelters, which do not differ significantly from each other.

Operations expenditure follows a narrower pattern: only medium shelters have an individually significant increase (\$8,600 per \$100,000 in grants), and this estimate differs significantly from small shelters ($p = 0.024$) but not from large shelters ($p = 0.186$), while small and large shelters are not distinguishable from each other ($p = 0.232$). I read this as a more limited form of heterogeneity, isolated to medium shelters rather than a consistent gradient across the size distribution.

Staffing

Volunteer count is the only staffing outcome with significant size heterogeneity after correction ($p = 0.024$); employee count does not ($p = 0.588$). Large shelters have a marginally significant decline in volunteer count (6.9 fewer volunteers per \$100,000 in grants), while small and medium shelters have no individually significant effect. The only significant pair-

wise difference is between small and large shelters ($p = 0.003$); medium shelters are not significantly distinguished from either small or large shelters. Given that this result rests on a single pairwise comparison and an individually marginal coefficient, I treat it as suggestive rather than a robust size-based pattern in volunteer engagement.

Violence: Hospital Discharges

The joint test rejects equal effects by size for both IPV and DV discharge rates after correction ($p = 0.041$ and $p = 0.049$, respectively). For IPV discharges, small and medium shelters are associated with a decline in the discharge rate, while large shelters are associated with an increase. Large shelters differ significantly from both small ($p = 0.005$) and medium ($p = 0.026$) shelters; small and medium are not distinguishable from each other ($p = 0.469$).

DV discharges follow the same directional pattern: a decline for small and medium shelters, an increase for large shelters. The pairwise structure is also similar (large vs. small $p = 0.030$; large vs. medium $p = 0.038$; small vs. medium $p = 0.984$), though none of the three tercile-specific DV discharge coefficients is individually significant on its own. The lagged IPV discharge coefficient for large shelters is also marginally significant, though this falls outside the corrected joint test and I do not treat it as confirmatory.

This pattern does not appear in the female assault discharge measure. Given the consistency of the small-versus-large split across IPV and DV discharges, but its absence elsewhere, I read this as a notable but exploratory finding rather than a robust conclusion about how shelter size shapes community-level effects.

Violence: Homicides and Child Maltreatment

No outcome in the UCR homicide or NCANDS panels rejects equality across shelter size after correction. Female and child homicide rates have raw joint-test p -values of 0.101 and 0.058, respectively; neither survives correction within the four-outcome UCR homicide family. NCANDS DV risk factor and substantiated reports for children ages 11–18 similarly

have raw p -values below 0.10 (0.020 and 0.075) but do not survive correction within the seven-outcome NCANDS family. I find no evidence that shelter size shapes the relationship between grant funding and either homicide rates or child maltreatment reporting.

6 Robustness

Table 16 reports two checks on the main financial and staffing results. First, I re-estimate the main specification after dropping each of the 46 shelters one at a time and report the median, minimum, and maximum coefficient across the 46 resulting estimates alongside the main estimate. For all five outcomes shown, the median leave-one-out estimate is nearly identical to the main estimate, and the minimum and maximum values remain close to the main estimate. This indicates that no single shelter is responsible for any of the five estimated effects. Full leave-one-out results for all revenue, expenditure, and staffing outcomes are reported in Appendix F.

Second, I restrict the sample to the 24 shelters with December fiscal year ends, the largest single fiscal-year group in the panel, to assess whether the rule used to assign Form 990 filings to federal award years introduces measurement error for shelters with non-December fiscal years. Total revenue, total expenditure, compensation, and employee count remain significant in the December FY subsample and are directionally consistent with the full-sample estimates, though the magnitudes differ: total revenue is roughly 40 percent larger in the December FY subsample (\$86,700 vs. \$61,900), while total expenditure, compensation, and employee count are 10 to 50 percent larger. Given that this restriction cuts the sample roughly in half, these magnitude differences are consistent with sampling variation, and I do not draw a strong conclusion from them.

The private donations result does not replicate in this subsample: the coefficient falls from $-\$38,900$ in the full sample, significant at the 5 percent level, to $-\$8,600$ in the December FY subsample, which is not statistically distinguishable from zero. This sensitivity

is consistent with the heterogeneity documented in Section 5.4, where private donations already showed a significant size-based pattern. One possibility is that the December FY sensitivity reflects a related timing issue: shelters can carry grant funds across fiscal-year boundaries, which is why the lagged-grant specification is used throughout this paper. If the timing of a private-donation response to a grant also varies with a shelter’s fiscal-year end, restricting to December filers could capture a different point in that response window than the full sample does. It remains an open question for the crowd-out result specifically, rather than for the shelter finance and staffing results more broadly.

Table 16: Consolidated Robustness: Leave-One-Out and December Fiscal Year Restriction

	Main Estimate	Leave-One-Out (46 samples)			December FY Only
		Median	Min	Max	
Total Revenue	61,900***	61,661	57,167	71,976	86,741***
Total Expenditure	57,300***	57,222	52,715	60,264	63,646***
Compensation	31,600***	31,546	28,309	33,408	35,503***
Employee Count	0.411***	0.377	0.271	0.442	0.621**
Private Donations	−38,900**	−39,151	−43,500	−28,152	−8,576

Notes: Leave-one-out columns report the main estimate alongside the median, minimum, and maximum coefficient obtained by re-estimating the specification after dropping each of the 46 shelters one at a time. December FY column restricts the sample to the 24 shelters with December fiscal year ends. The private donations coefficient in the December FY subsample is not statistically significant. Full leave-one-out results for all revenue, expenditure, and staffing outcomes, and the full December FY comparison, are reported in Appendix F. All specifications include shelter and year fixed effects with standard errors clustered at the shelter level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

7 Conclusion

Policymakers allocating federal resources to domestic violence shelter programs have lacked an empirical basis for evaluating the effects of that funding. This paper offers the first such evidence. Using a novel panel of Georgia DV shelters linked to government grant records,

organizational finances, and county-level outcomes, I trace the relationship between federal funding, shelter operations, and community-level outcomes.

A \$100,000 increase in government grants raises shelter revenue by \$61,900. Private donations decline by \$38,900 per \$100,000 in grants, a 13 percent decline relative to the sample average. This is a smaller degree of crowd-out than the 70 to 75 cents per dollar found elsewhere in the nonprofit literature. Total expenditure rises by \$57,300, directed primarily toward compensation and program services. Grants also raise employee counts, by 0.411 employees per \$100,000 in additional grants. I also find a marginally significant increase in compensation per employee, suggesting that grants raise average pay in addition to headcount. These estimates are stable across leave-one-out samples and a subsample restricted to shelters with a December fiscal year, with one exception: the private donations estimate is not statistically distinguishable from zero in the December fiscal year subsample. This sensitivity is consistent with the size heterogeneity documented in Section 5.4, where private donations already showed a significant size-based pattern. A lagged grant specification indicates that shelters spend a meaningful share of grant dollars in the fiscal year following receipt, so the contemporaneous expenditure and compensation estimates are best read as an upper bound on the within-year spending response rather than a complete accounting of how grant dollars are spent.

Georgia shelters already offer a broad range of services: fourteen of thirty-nine services recorded in the CJCC data are offered by every reporting shelter at some point during 2018–2022 (Section 3.7). One interpretation of the staffing and compensation results, taken together with this existing service breadth, is that additional funding increases the intensity of existing services – for example, through more staff time devoted to each client – rather than expanding the range of services offered. The CJCC victim-count data cannot distinguish this possibility from a scenario in which the same number of victims are served without any change in the intensity of service each receives, since these data record unique victims served rather than the duration or intensity of service provided. I am unable to test this interpretation

directly with the data available.

Estimated effects on community-level violence and child maltreatment are small and inconsistent across specifications. The one exception, a small increase in female assault hospital discharges, does not correspond to a broader pattern across related outcomes, and heterogeneity by shelter size is similarly limited to hospital discharges rather than homicides or child maltreatment. Taken together, these results do not point clearly toward any single channel through which shelter funding could affect community-level violence, and I do not find evidence of a stable, corroborated effect on the incidence of domestic violence or child maltreatment at the county level.

A minimal aggregate effect on incidence is still consistent with funding impacting DV victims. Funding may improve the quality of services provided to victims already in contact with a shelter without expanding the number of victims served or reached, and county-level incidence data cannot detect a pure quality effect: a victim who receives more intensive safety planning or advocacy, but whose decision to leave or whose contact with law enforcement or hospitals does not change, would not appear as a change in aggregate violence counts. Evaluating whether funding improves outcomes for victims already receiving services would require individual-level data on service recipients that does not exist in any of the sources used in this paper.

This intensive-margin interpretation does not, however, fully resolve the aggregate null result. Domestic violence is frequently chronic and repeated within a relationship, so if improved services meaningfully help victims exit abusive situations or reduce their risk of future violence, this should eventually appear as fewer repeat incidents at the county level, not only as unobserved improvements in service quality. This argument is complicated by help-seeking behavior, however: funding that connects more victims to medical care or law enforcement could increase measured incidents even as underlying violence declines, offsetting or masking a true improvement. The absence of a detectable aggregate effect is therefore not fully conclusive in either direction: it is consistent with quality improvements that do

not translate into fewer incidents, but also with a true decline that is offset by increased help-seeking among victims already in contact with services.

Domestic violence shelters are the primary resource available to victims outside the criminal justice system. This paper establishes that government grants increase shelter revenue, expenditure, and staffing, with evidence of crowd-out concentrated in private donations. Community-level violence and child maltreatment outcomes show small and largely uncorroborated effects, providing limited evidence that funding reduces the incidence of domestic violence or child maltreatment at the county level. Whether funding nonetheless improves conditions for the victims a shelter directly serves is a question this paper's data cannot answer, and the county-level null result does not resolve it: it is consistent both with funding having little effect on victims already in contact with services, and with a true improvement that is offset by increased help-seeking. This paper provides the data infrastructure and the empirical foundation for the next stage of that question.

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A DV Search Terms

The following terms were used to flag potential DV service providers in the first stage of shelter identification. Terms were derived from a review of Form 990 filings from shelters identified through an independent online search, and were selected to capture the range of terminology used by organizations to describe domestic violence programming.

Table 17: Search Terms Used for DV Shelter Identification

<i>Search Terms</i>	
domestic violence	intimate partner abuse
domestic abuse	battered women
intimate partner violence	domestic assault
family violence	family abuse
spousal abuse	spouse abuse
dating violence	DV
relationship violence	IPV

B DV Shelter Database Variables

Table 18: IRS 990 Database Variable Identification

Database Variable	Level	990 Entry and Instruction
Organization Name	Organization	Name of organization
Address	Organization	City or town, state or province, country, and ZIP
EIN	Organization	Employer Identification Number
Employee Count	Organization	Total number of individuals employed in calendar year
Volunteer Count	Organization	Total number of volunteers (estimate if necessary)
Mission Statement	Organization	Briefly describe the organization's mission
Tax Year Start	Organization	Tax year beginning on Month - Day - Year
Tax Year End	Organization	Tax year ending on Month - Day - Year
Total Revenue	Organization	Total revenue
Total Expenditure	Organization	Total expenses
Total Compensation	Organization	Salaries, other compensation, employee benefits
Program Description‡	Program	Describe the organization's program service accomplishments of each of its three largest program services, as measured by expenses
Program Expenditure‡	Program	Report the total expenses for each program service
Officers, Directors, Trustees, Key Employees		
Employee Name§	Organization	Name
Employee Title§	Organization	Title
Employee Compensation§	Organization	Reportable compensation from the organization
Statement of Revenue (Organization Level)		
Program Revenue	Organization	Program Revenue
Federated Campaigns Revenue	Organization	Federated Campaigns
Membership Dues Revenue	Organization	Membership Dues
Related Organizations Revenue	Organization	Related Organizations
Fundraising Events Revenue	Organization	Fundraising Event
Government Grants Revenue	Organization	Government Grants
All Other Contributions	Organization	All other contributions, gifts, grants, and similar amounts not included above
Statement of Functional Expenses (Organization Level)		
Fundraising Expenditure	Organization	Total fundraising expenses
Occupancy Expenditure	Organization	Facilities expenses including rent, utilities, property insurance, real estate taxes, and mortgage interest
Legal Expenditure	Organization	Legal fees for services (non-employees)
Accounting Expenditure	Organization	Accounting fees for services (non-employees)
Management Fee Expenditure	Organization	Management fees for services (non-employees)
Lobbying Expenditure	Organization	Lobbying fees for services (non-employees)
Other Fee-for-Service	Organization	Any other fees for services (non-employees)
Advertising Expenditure	Organization	Advertising and promotion costs
Office Expenditure	Organization	Office supplies, equipment, and postage
Travel Expenditure	Organization	Travel and transportation costs
Conference Expenditure	Organization	Conferences, conventions, and meetings
Employee Benefits Expenditure	Organization	Employee benefits (not including pension plan contributions)
Payroll Taxes Expenditure	Organization	Payroll taxes
IT Expenditure	Organization	Information Technology expenses
Interest	Organization	Total interest expenses
Depreciation	Organization	Depreciation, depletion, and amortization

Notes: The table describes the variables in the DV Shelter Database and their respective 990 form entry descriptions. Organization-level variables are reported once per organization per year. ‡ Program-level variables are reported up to three times per organization per year, once for each of the organization's three largest program services by expense. § These organization-level variables are reported up to seventeen times per organization per year, once for each officer, director, trustee, or key employee.

Table 19: Constructed Revenue and Expenditure Variables

Variable	Construction
Expenditure	
Fee-for-Service Expenditure	Sum of management, legal, accounting, lobbying, fundraising, and other fee-for-service expenditures
Operations Expenditure	Sum of advertising, office, information technology, travel, and conference expenditures
Other Expenditure	Total expenditure minus compensation, fee-for-service expenditure, operations expenditure, occupancy expenditure, and fundraising expenditure
Revenue	
Private Donations	Sum of all other contributions and fundraising event revenue
Private Donation Share	Private donations divided by total revenue
Other Revenue	Total revenue minus government grants, private donations, and program service revenue
Other Revenue Share	Other revenue divided by total revenue
Program Revenue Share	Program service revenue divided by total revenue

Notes: This table describes revenue and expenditure variables constructed from the raw IRS Form 990 entries listed in Table 18. Fee-for-service, operations, and other expenditure categories are constructed to be mutually exclusive and collectively exhaustive of total expenditure. Private donations, other revenue, and program service revenue are constructed to be mutually exclusive and collectively exhaustive of total revenue, alongside government grants.

C Characterizing the Grant Allocation Process

This appendix characterizes how government grant funding was allocated across shelters and over time, supporting the parallel trends discussion in Section 4.2. Different allocation mechanisms imply different empirical patterns in the data. I examine whether grant funding reflects permanent differences across shelters or common year-to-year variation, whether the 2016 funding expansion scaled up existing allocations or reshuffled them, whether observable shelter and county characteristics predict grant levels or grant changes, and how persistent and rank-stable funding is over the panel.

C.1 Did the Funding Expansion Scale Up Existing Allocations or Reshuffle Funding?

Figure 5 plots grant funding in 2016, 2018, 2019, and 2023 against 2013 grant funding, together with the 45-degree line and a fitted regression line from $G_{it} = \alpha + \rho G_{i,2013} + \varepsilon_{it}$;

Table 20: Full Service Provision Breakdown, 2018–2022

Service	Shelter-Yr (%)	Ever (%)	Always (%)	Mean Victims/Yr
TANF/Government Benefits Assistance	87.4	100.0	63.0	110
Criminal Justice Support/Advocacy	90.4	100.0	69.6	327
Crisis Coordination	87.0	100.0	60.9	2,581
Crisis Line	99.1	100.0	95.7	2,017
Emergency Financial Assistance	91.3	100.0	69.6	178
Emergency Legal Assistance	90.4	100.0	73.9	801
Individual Support	99.1	100.0	95.7	7,599
Interagency Advocacy	90.9	100.0	76.1	823
Non-Emergency Legal Assistance	92.2	100.0	73.9	523
Referral to Services, Support, Resources	99.1	100.0	95.7	1,107
Relocation Assistance	90.0	100.0	67.4	210
Safe Housing	97.8	100.0	91.3	3,684
Support Group	93.9	100.0	84.8	617
Transportation Assistance	98.7	100.0	93.5	1,297
Child/Dependent Care Assistance	82.6	97.8	56.5	259
Information on Criminal Justice Process	72.2	97.8	41.3	170
Provide Info Explaining Victim Rights	76.5	95.7	45.7	186
Assistance Applying for Victims Compensation	73.9	95.7	45.7	40
Provide Access to Language Assistance	63.5	95.7	34.8	78
Advocacy/Accompaniment–Emergency Medical	60.9	93.5	28.3	43
Non-Traditional Therapeutic Intervention	66.5	93.5	39.1	340
Therapy from Licensed Clinician/Counselor	65.7	93.5	39.1	200
Civil Legal Attorney Assistance–TPO	65.2	87.0	30.4	96
Assistance Placing Animals in Distress	55.2	82.6	21.7	30
Law Enforcement Interview Advocacy	42.6	78.3	15.2	30
Civil Legal Attorney Assistance–Family Law	38.3	71.7	6.5	47
Transitional Housing	39.1	71.7	17.4	1,147
Immigration Assistance–Advocate	33.0	69.6	10.9	27
Victim Notification of Criminal Justice Events	33.5	67.4	8.7	46
Group Therapy	24.8	58.7	4.3	62
Victim Impact Assistance	16.5	47.8	2.2	22
Assistance with Restitution	7.8	28.3	0.0	0.75
Prosecution Interview Advocacy	13.9	32.6	2.2	12.4
On-Scene Crisis Response	15.7	34.8	2.2	4.9
Performance of Initial Forensic Medical Exam	13.0	19.6	8.7	9.4
Performance of Forensic Interview	13.9	17.4	8.7	8.8
Performance of Follow-Up Forensic Medical Exam	10.9	17.4	2.2	0.67
Immigration Assistance–Attorney	5.2	13.0	0.0	6.9

Notes: Sample includes 46 shelters with CJCC service records, 2018–2022 ($N = 230$ shelter-years unless otherwise noted). Shelter-Yr is the shelter-year average offering rate. Ever is the share of the 46 shelters offering the service in at least one sample year. Always is the share offering it in every year observed. Mean victims/year is the shelter-year average. Sorted by Ever (descending) within the near-universal/non-near-universal groupings implied in Table 4. Excludes “Other Service Provided,” which no shelter reports offering. Data come from CJCC utilization reports.

fitted slopes appear in Table 21. The estimated slope is 0.808 in 2016, 1.059 in 2018, and 0.960 in 2019, with each fitted line lying close to or above the 45-degree line. A shelter's rank in the 2013 funding distribution therefore predicts its rank through 2019 closely, while the typical shelter received more funding in later years than in 2013 — a pattern consistent with the funding expansion scaling up existing allocations rather than reallocating funding toward previously low-funded shelters. By 2023, the estimated slope rises to 1.342, indicating that shelters with larger 2013 baseline funding received disproportionately larger increases by the end of the panel.

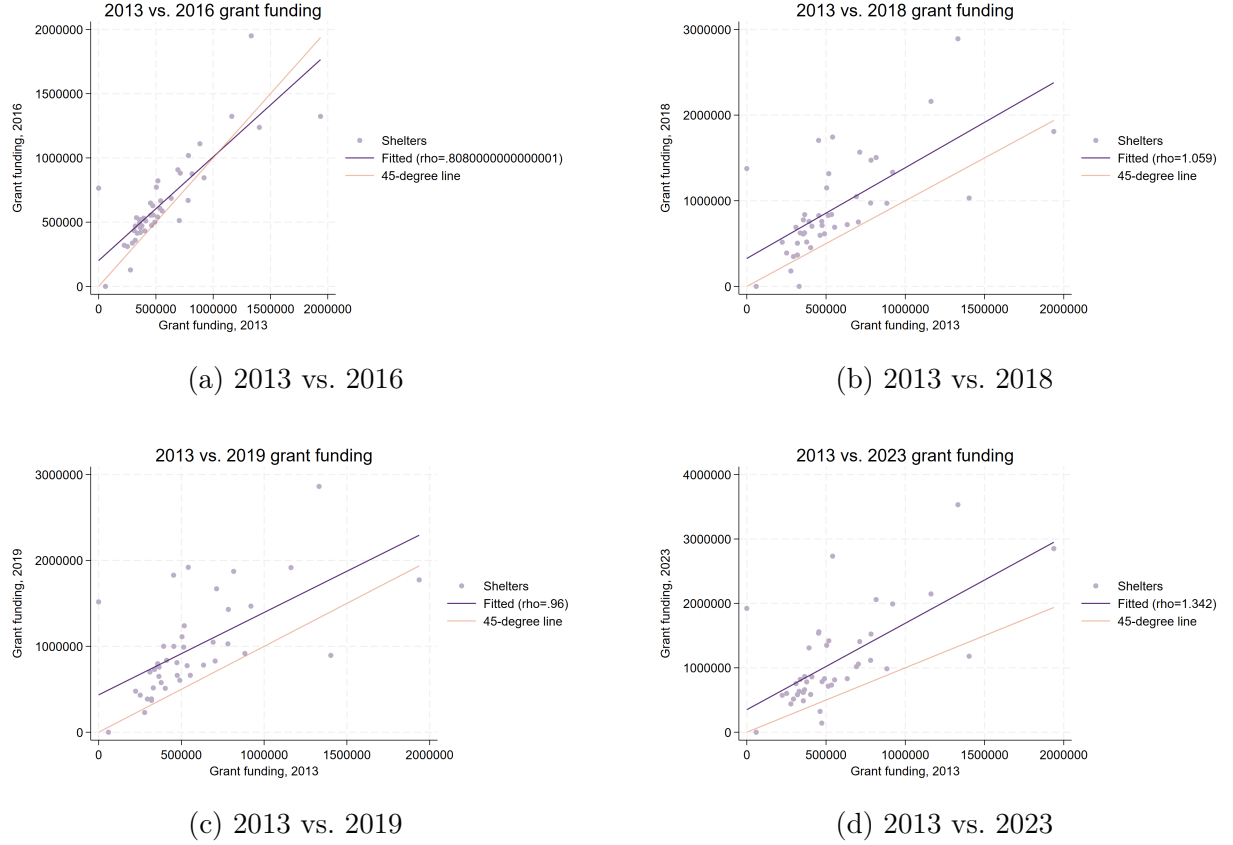


Figure 5: Grant Funding in Later Years versus 2013 Baseline Funding

C.2 Post-Expansion Grant Growth and Baseline Funding

Table 22 regresses several measures of post-expansion grant growth on 2013 baseline funding: the initial change from 2015 to 2016, the average post-period increase relative to the average

Table 21: Fitted Slopes of Grant Funding on 2013 Baseline Funding

Year	$\hat{\rho}$
2016	0.808
2018	1.059
2019	0.960
2023	1.342

Slope from $G_{it} = \alpha + \rho G_{i,2013} + \varepsilon_{it}$, estimated separately by year.

pre-period level, the same measure using the 2013–2015 average as the baseline regressor, and cumulative post-expansion funding growth relative to a counterfactual continuation of 2013–2015 average funding. The coefficient on baseline funding is positive in each specification, ranging from 0.076 to 0.718, but none is statistically distinguishable from zero. Given the small number of shelters underlying these cross-sectional regressions, this specification has limited power to detect a relationship between baseline funding and subsequent growth even if one is present.

Table 22: Post-Expansion Grant Growth on Baseline (2013) Funding

	(1) Initial (16–15)	(2) Avg. post-pre	(3) Avg. post – 3yr avg. pre	(4) Cum. growth
Grant, 2013	0.0759 (0.0889)	0.188 (0.182)		0.718 (0.728)
Grant, 2013–15 avg.			0.316 (0.222)	
Constant	41802.7 (39971.0)	165046.7 (90284.5)	99188.1 (105697.3)	703075.1* (367972.3)
N	42	44	45	42

Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.3 The Annual Grant Allocation Equation

To assess whether observable measures of shelter need predict grant funding, I estimate $G_{it} = \alpha_i + \gamma_t + X'_{i,t-1}\beta + \varepsilon_{it}$ using the 2014–2023 panel, where $X_{i,t-1}$ includes lagged shelter finances, lagged county violence, lagged county demographic characteristics, and lagged

shelter staffing, grouped into conceptually meaningful blocks. Table 23 reports results by block, a combined specification including all blocks simultaneously, and a first-differenced specification using the annual grant change as the outcome.

Among the four blocks, the finance block is the most predictive ($F = 118.71$, $p < 0.001$, within $R^2 = 0.643$), though this partly reflects an accounting identity: grant funding is itself a component of total revenue, and expenditure is mechanically linked to revenue through the shelter’s budget constraint. The staffing block is also jointly significant ($F = 5.80$, $p = 0.007$), with employee count individually predictive. County demographics are jointly significant as well ($F = 4.59$, $p = 0.002$), but explain little of the variation (within $R^2 = 0.126$) and are driven primarily by county birth counts. County violence is not informative on its own. In the fully saturated combined specification, the coefficient on lagged grant funding falls to 0.141 and loses significance ($p = 0.586$), reflecting collinearity between grant funding, revenue, and expenditure.

The first-differenced specification tells a different story: none of the lagged shelter finance, county violence, or county demographic variables predicts the size of a shelter’s grant *change*, and within R^2 falls to 0.151. Grant *levels* are therefore highly predictable from a shelter’s own financial history, largely mechanically, while grant *changes* are not predicted by any observable measure tested here — a pattern inconsistent with an allocation process that adjusts funding annually in response to observable shelter need.

C.4 Do Prior Outcome Changes Predict Future Grant Changes?

I next test directly whether deteriorating or improving shelter conditions predict the following year’s grant change, estimating $\Delta G_{it} = \gamma_t + \beta \Delta Y_{i,t-1} + \alpha_i + \varepsilon_{it}$ for twelve outcomes. Table 24 reports these results. Two coefficients are significant at conventional levels: a prior increase in the female homicide rate predicts a smaller next-period grant change ($\hat{\beta} = -1,791.4$, $p = 0.018$), and a prior increase in the DV homicide rate is marginally significant ($\hat{\beta} = -8,639.6$, $p = 0.052$). Both signs are inconsistent with a reactive, needs-based allocation process, under

Table 23: Blocks of Lagged Predictors and the Annual Grant Allocation Equation

	(1) Finance block	(2) Violence block	(3) Demographics block	(4) Combined (levels)	(5) Δ Grant
Lagged grant funding				0.141 (0.258)	
Lagged total revenue	0.249*** (0.070)			0.149 (0.200)	-0.165 (0.115)
Lagged total expenditure	0.366** (0.139)			0.421*** (0.097)	0.048 (0.068)
Lagged compensation	0.267 (0.193)			0.190 (0.204)	0.262 (0.153)
Lagged private donations	-0.165** (0.070)			-0.078 (0.181)	0.112 (0.115)
Lagged program revenue	0.312 (0.261)			-0.382 (0.262)	0.022 (0.174)
Lagged operating margin	omitted			omitted	omitted
Lagged DV homicide rate		2993.8 (8668.9)		-9354.4 (5803.6)	-8483.3 (5598.4)
Lagged county births			-665.7*** (195.2)	-229.4** (101.8)	-1.1 (6.5)
Lagged county pc income			35.1 (25.6)	17.9* (9.7)	2.9 (2.0)
Lagged county med. HH income			-7.2 (9.3)	1.5 (6.5)	-2.1 (1.7)
Lagged county poverty			-2034.6 (10783.8)	9419.6 (6211.7)	-1062.4 (2761.9)
Lagged county unemployment			-9649.2 (34170.4)	-12765.7 (20535.5)	12911.1 (12843.0)
Joint test (F)	118.71***	0.12	4.59***	152.85***	
Within R^2	0.643	0.000	0.126	0.628	0.151
N	442	423	312	306	306

Notes: Shelter and year fixed effects included in columns (1)–(4); year fixed effects only in column (5), the first-differenced specification. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

which a worsening homicide rate would predict a larger grant change. Applying a Benjamini-Hochberg correction across the twelve tests (Benjamini & Hochberg, 1995), neither result remains significant (adjusted $p = 0.213$ and $p = 0.250$).

Table 24: Do Prior Changes in Shelter Outcomes Predict the Next Grant Change?

Prior change in:	Coefficient	SE	p	p (Bonferroni)	p (BH)
Total revenue	-0.052	(0.035)	0.142	1.000	0.340
Total expenditure	-0.035	(0.100)	0.725	1.000	0.791
Compensation	0.219	(0.123)	0.083	0.998	0.250
Employee count	722.6	(1,417.0)	0.613	1.000	0.791
Private donations	0.019	(0.051)	0.711	1.000	0.791
IPV hospital discharges	1,483.9	(1,457.2)	0.314	1.000	0.628
DV hospital discharges	49.6	(784.0)	0.950	0.950	0.950
Female assault discharges	-80.3	(45.0)	0.081	0.977	0.250
DV homicide rate	-8,639.6	(4,330.4)	0.052	0.629	0.250
Family violence homicide rate	-3,060.5	(3,695.2)	0.412	1.000	0.706
Female homicide rate	-1,791.4	(726.8)	0.018	0.213	0.213
Substantiated NCANDS reports	-31.4	(55.6)	0.576	1.000	0.791

Notes: Dependent variable is the annual grant change, ΔG_{it} . Each row is a separate regression of the grant change on the prior-year change in the listed outcome, with shelter and year fixed effects and standard errors clustered by county. p (Bonferroni) and p (BH) report Bonferroni-corrected (Dunn, 1961) and Benjamini-Hochberg-adjusted (Benjamini & Hochberg, 1995) p -values across the twelve tests.

C.5 Pre-Expansion Trends and Post-Expansion Growth

To assess whether shelters already on a particular funding trajectory before 2016 received larger subsequent increases, I recover each shelter’s linear pre-2016 grant trend, $\hat{b}_i$, from $G_{it} = a_i + b_i t + u_{it}$ estimated over 2013–2015, and regress post-expansion growth on this trend. Table 25 reports the result: the estimated coefficient is positive ($\hat{\theta} = 0.551$) but not statistically distinguishable from zero ($p = 0.187$). Because the pre-period spans only three years, this estimate is imprecise and should be read as inconclusive rather than as evidence against a relationship between pre-existing trends and subsequent funding growth.

Table 25: Pre-2016 Grant Trend and Post-Expansion Funding Growth

	$\Delta G_i^{post-pre}$
Pre-2016 grant trend, $\hat{b}_i$	0.551 (0.411)
Constant	268806.4*** (39402.7)
N	45

Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.6 Funding Persistence and Rank Stability

I estimate the persistence of grant funding using $G_{it} = \alpha_i + \gamma_t + \rho G_{i,t-1} + \varepsilon_{it}$ over 2014–2023. The estimated persistence coefficient is $\hat{\rho} = 0.745$ ($p < 0.001$), indicating that grant funding follows a persistent trajectory rather than a sequence of independent annual variation.

Table 26 reports quintile transition matrices comparing each shelter’s grant funding quintile in 2013 to its quintile in 2016, 2019, and 2023. Shelters in the bottom quintile of 2013 funding remain there 66.7 percent of the time by 2016, 77.8 percent by 2019, and 66.7 percent by 2023. Shelters in the top quintile remain there 87.5 percent of the time by 2016 and 62.5 percent by both 2019 and 2023. Middle-quintile shelters transition more frequently: no middle-quintile cell exceeds 44 percent retention in any period, and most transitions move a shelter to an adjacent quintile rather than across the full distribution. Funding rank is therefore most stable at the extremes of the distribution and least stable among shelters with moderate funding levels.

C.7 Summary

Across these exercises, grant funding is best characterized as highly persistent and primarily explained by permanent differences across shelters rather than by common year effects, observable shelter need, or county conditions. The 2016 funding expansion is consistent with a scaling up of existing allocations through 2019, with some divergence between high- and low-baseline shelters emerging by 2023. No measure of lagged shelter finances, county

Table 26: Grant Funding Quintile Transition Matrices

2013 quintile	2016 quintile					2019 quintile					2023 quintile				
	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
1	66.7	22.2	0.0	11.1	0.0	77.8	11.1	0.0	0.0	11.1	66.7	22.2	0.0	0.0	11.1
2	33.3	33.3	33.3	0.0	0.0	22.2	55.6	11.1	11.1	0.0	22.2	44.4	22.2	11.1	0.0
3	0.0	33.3	33.3	33.3	0.0	0.0	25.0	37.5	25.0	12.5	22.2	22.2	22.2	11.1	22.2
4	0.0	11.1	22.2	44.4	22.2	0.0	11.1	33.3	33.3	22.2	0.0	11.1	44.4	33.3	11.1
5	0.0	0.0	0.0	12.5	87.5	0.0	0.0	25.0	12.5	62.5	0.0	0.0	12.5	25.0	62.5

Cell entries are row percentages: the share of shelters in a given 2013 quintile that fall into each quintile in the comparison year. Quintile 1 is the lowest grant funding; quintile 5 is the highest.

violence, or county demographic characteristics predicts annual grant *changes*, even though grant *levels* are strongly related to a shelter’s own financial history. This evidence supports treating funding variation in the shelter-year panel as substantially disconnected from contemporaneous shelter need at an annual frequency, which bears directly on the identification strategy used in the main analysis.

D Poisson Specification

For the violence outcomes I additionally estimate a Poisson quasi-maximum likelihood (QMLE) specification:

$$\mathbb{E}[Y_{it} \mid \text{Grants}_{it}, P_{it}] = \exp(\beta \cdot (\text{Grants}/\$100\text{k})_{it} + \alpha_i + \gamma_t + \ln P_{it}), \quad (3)$$

where P_{it} is the relevant population, which enters as an exposure offset. The coefficient β is interpreted as the approximate percent change in the outcome rate per \$100,000 increase in grants. Poisson QMLE is well-suited for these outcomes because it is consistent under arbitrary distributions of integer-valued data, accommodates the preponderance of zero counts that characterizes the homicide outcomes, and imposes a natural log-linear relationship between grants and the outcome rate. Standard errors are clustered at the county level.

Table 27 reports the Poisson counterpart to Table 10. The lagged coefficient for female assault discharges is positive and significant at the 1 percent level, consistent with the OLS result: this is the same outcome that remains significant after correcting for multiple testing in the main text. The IPV and DV discharge coefficients are not significant at either lag, matching the OLS null result for these two outcomes.

Table 27: Effects of Government Grants on Hospital Discharges: Poisson

	(1)	(2)	(3)
	IPV Discharge	DV Discharge	Assault Discharge
	Count, pop.	Count, pop.	Count, pop.
	offset	offset	offset
Grants (\$100k)	0.003	0.012	-0.006
	(0.021)	(0.015)	(0.006)
Grants (lagged)	-0.016	-0.006	0.018***
	(0.021)	(0.016)	(0.006)
Observations	264	264	264

Notes: Poisson counterpart to Table 10. Each column regresses the underlying discharge count on grants, with population entered as an exposure offset via `ppmlhdfc`; the resulting coefficient approximates the percent change in the discharge rate per \$100,000 increase in grants. Column definitions match Table 10. Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 28 reports the Poisson counterpart to Table 11. The contemporaneous and lagged IPV homicide coefficients are both significant, with the same opposing signs found in the OLS specification: a decline in the same year and an increase of similar direction the following year. This agreement between OLS and Poisson does not resolve the interpretive difficulty raised in the main text; the sign reversal remains difficult to attribute to a single underlying mechanism regardless of estimator. The remaining homicide measures — DV, female, and child homicides — are not significant at either lag, consistent with the OLS results.

Table 28: Effects of Government Grants on DV Homicides: Poisson

	(1)	(2)	(3)	(4)
	IPV Homicide	DV Homicide	Female Homicide	Child Homicide
Grants (\$100k)	-0.093*** (0.033)	-0.015 (0.049)	-0.051 (0.056)	-0.022 (0.035)
Grants (lagged)	0.121** (0.050)	0.025 (0.051)	-0.033 (0.059)	-0.011 (0.043)
Observations	180	186	152	139

Notes: Poisson counterpart to Table 11. Each column regresses the underlying homicide count on grants, with population entered as an exposure offset via `ppmlhdfe`; the resulting coefficient approximates the percent change in the homicide rate per \$100,000 increase in grants. Sample sizes differ from Table 11 because `ppmlhdfe` drops counties with no within-group outcome variation. Column definitions match Table 11. Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 29 reports the Poisson counterpart to Table 12. None of the seven outcomes is significant at either lag. This includes substantiated reports for children ages 11–18, which showed a marginally significant contemporaneous coefficient in the OLS specification before failing to survive correction for multiple testing; the absence of even marginal significance in the Poisson specification is consistent with treating that OLS result as noise rather than a genuine effect.

Table 29: Effects of Government Grants on Child Maltreatment Reports: Poisson

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	All Ages	Infant	Ages 1–3	Ages 4–10	Ages 11–18	DV Risk	Foster
	Substantiated Reports					Factor	Services
Grants (\$100k)	-0.005	-0.002	-0.004	-0.002	-0.010	0.007	0.002
	(0.009)	(0.010)	(0.012)	(0.012)	(0.008)	(0.008)	(0.008)
Grants (lagged)	0.003	0.003	-0.003	-0.002	-0.008	0.011	0.002
	(0.015)	(0.010)	(0.016)	(0.016)	(0.015)	(0.011)	(0.011)
Observations	228	224	224	224	224	224	224

Notes: Poisson counterpart to Table 12. Each column regresses the underlying report count on grants, with population under 18 entered as an exposure offset via `ppmlhdfc`; the resulting coefficient approximates the percent change in the report rate per \$100,000 increase in grants. Sample sizes differ from Table 12 because `ppmlhdfc` drops counties with no within-group outcome variation. Column definitions match Table 12. Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

E Full Heterogeneity by Shelter Size

This appendix reports tercile-specific coefficients for every outcome examined in the shelter-size heterogeneity analysis in Section 5.4. Tables 30 through 32 report revenue, expenditure, and staffing outcomes at the shelter level. Tables 33 through 35 report hospital discharge, homicide, and child maltreatment outcomes at the county level, aggregating shelter grant funding within each county-year as described in Section 4. Table 15 in the main text reports only the nine outcomes with statistically significant heterogeneity after correction; the remaining outcomes are reported here for completeness.

Table 30: Effect of Government Grants on Revenue Components by Shelter Size

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Revenue Level (\$100k)				Share of Total Revenue			
	Total	Private	Program	Other	Grants	Private	Program	Other
	Revenue	Donations	Revenue	Revenue		Donations	Revenue	Revenue
<i>Panel A: Small Shelters</i>								
Grants (\$100k)	0.590***	-0.317**	0.012	-0.105*	0.066***	-0.044*	0.002	-0.024**
	(0.214)	(0.160)	(0.009)	(0.055)	(0.021)	(0.026)	(0.001)	(0.011)
Grants (lagged)	-0.061	0.021	-0.008	-0.075	0.004	0.003	-0.001	-0.006
	(0.055)	(0.074)	(0.005)	(0.063)	(0.005)	(0.008)	(0.001)	(0.007)
Sample Mean	6.890	1.333	0.049	0.325	0.717	0.217	0.009	0.057
N	176	168	170	168	168	168	170	168
<i>Panel B: Medium Shelters</i>								
Grants (\$100k)	1.081***	0.039	-0.002	0.043	0.014***	-0.015*	0.000	0.000
	(0.053)	(0.066)	(0.025)	(0.029)	(0.005)	(0.008)	(0.003)	(0.003)
Grants (lagged)	0.005	0.032	-0.005	-0.021	-0.001	0.003	-0.000	-0.001
	(0.027)	(0.024)	(0.005)	(0.019)	(0.001)	(0.002)	(0.000)	(0.001)
Sample Mean	10.777	2.283	0.376	0.778	0.675	0.209	0.040	0.075
N	165	161	165	161	161	161	165	161
<i>Panel C: Large Shelters</i>								
Grants (\$100k)	0.798***	-0.193	0.028	-0.037	0.023***	-0.018***	-0.001	-0.004
	(0.114)	(0.124)	(0.036)	(0.049)	(0.005)	(0.005)	(0.002)	(0.005)
Grants (lagged)	-0.084	-0.070	-0.025*	0.011	0.002	-0.001	-0.001	-0.000
	(0.086)	(0.077)	(0.013)	(0.014)	(0.002)	(0.002)	(0.001)	(0.001)
Sample Mean	19.792	5.232	0.374	1.459	0.649	0.244	0.017	0.090
N	165	165	165	165	165	165	165	165

Notes: The table shows the estimated effects of a \$100,000 increase in government grants, adding a one-year lagged grant term to the specification, estimated separately by baseline shelter size. Size is defined by 2013 revenue terciles. Observations are at the shelter-fiscal-year level from 2013–2023. All regressions include shelter and year fixed effects, estimated separately within each tercile. Private donations comprise fundraising event revenue and direct contributions (IRS Form 990 Part VIII lines 1c and 1f). Program revenue is reported on Part I of Form 990. Other revenue includes federated campaigns, membership dues, grants from other organizations, and miscellaneous receipts. Data come from the DV Shelter Database. Standard errors in parentheses are clustered at the shelter level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 31: Effect of Government Grants on Expenditure Components by Shelter Size

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Primary	Compensation	Contract	Operations	Occupancy	Fundraising	Other
	Expenditure	Program		Work			Expenditure	
	(\$100k)							
Panel A: Small Shelters								
Grants (\$100k)	0.404**	0.398***	0.245*	0.012*	0.009	0.036***	-0.003	0.106*
	(0.206)	(0.112)	(0.133)	(0.007)	(0.007)	(0.012)	(0.006)	(0.064)
Grants (lagged)	-0.020	0.082***	-0.014	0.002	0.001	0.001	0.001	-0.011
	(0.049)	(0.029)	(0.025)	(0.003)	(0.007)	(0.007)	(0.002)	(0.029)
Sample Mean	6.459	4.860	3.994	0.145	0.220	0.385	0.079	1.579
N	176	168	176	176	176	176	168	168
Panel B: Medium Shelters								
Grants (\$100k)	0.959***	0.825***	0.524***	-0.002	0.086***	0.057***	0.004	0.289***
	(0.045)	(0.082)	(0.069)	(0.013)	(0.020)	(0.020)	(0.007)	(0.087)
Grants (lagged)	0.032	-0.014	0.072**	0.003	0.003	0.008**	-0.004	-0.050**
	(0.024)	(0.035)	(0.031)	(0.004)	(0.009)	(0.004)	(0.005)	(0.021)
Sample Mean	10.132	7.861	6.130	0.243	0.390	0.613	0.376	2.353
N	165	161	165	165	165	165	161	161
Panel C: Large Shelters								
Grants (\$100k)	0.696***	0.562***	0.335***	0.127**	0.016	-0.002	0.085	0.135
	(0.119)	(0.110)	(0.058)	(0.050)	(0.020)	(0.013)	(0.052)	(0.111)
Grants (lagged)	-0.053	-0.003	-0.063**	0.025	-0.002	-0.013**	0.040	-0.040
	(0.039)	(0.035)	(0.032)	(0.021)	(0.005)	(0.005)	(0.041)	(0.050)
Sample Mean	18.215	13.653	10.185	1.093	0.719	0.912	1.043	4.264
N	165	165	165	165	165	165	165	165

Notes: The table shows the estimated effects of a \$100,000 increase in government grants, adding a one-year lagged grant term to the specification, estimated separately by baseline shelter size. Size is defined by 2013 revenue terciles. Observations are at the shelter-fiscal-year level from 2013–2023. All regressions include shelter and year fixed effects, estimated separately within each tercile. Compensation includes all W-2 employee compensation. Contract work comprises all fee-for-service expenditure lines. Operations includes office, information technology, advertising, travel, and conference expenditures. Data come from the DV Shelter Database. Standard errors in parentheses are clustered at the shelter level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 32: Effect of Government Grants on Staffing Outcomes by Shelter Size

	(1)	(2)	(3)	(4)	(5)
	Compensation	Highest Paid	Contract	Employee	Volunteer
	(\$100k)	Employee (\$100k)	Work (\$100k)	Count	Count
<i>Panel A: Small Shelters</i>					
Grants (\$100k)	0.245*	0.020***	0.012*	0.920***	-1.628
	(0.133)	(0.008)	(0.007)	(0.262)	(1.476)
Grants (lagged)	-0.014	0.014***	0.002	0.132	0.488
	(0.025)	(0.005)	(0.003)	(0.123)	(0.734)
Sample Mean	3.994	0.494	0.145	16.29	31.57
N	176	89	176	168	106
<i>Panel B: Medium Shelters</i>					
Grants (\$100k)	0.524***	-0.006	-0.002	1.074***	-0.307
	(0.069)	(0.007)	(0.013)	(0.278)	(3.966)
Grants (lagged)	0.072**	0.002	0.003	0.030	-0.272
	(0.031)	(0.002)	(0.004)	(0.099)	(1.242)
Sample Mean	6.130	0.464	0.243	29.75	65.00
N	165	66	165	161	104
<i>Panel C: Large Shelters</i>					
Grants (\$100k)	0.335***	-0.014	0.127**	0.489**	-6.919*
	(0.058)	(0.012)	(0.050)	(0.209)	(3.595)
Grants (lagged)	-0.063**	0.001	0.025	-0.196**	-0.061
	(0.032)	(0.004)	(0.021)	(0.094)	(1.348)
Sample Mean	10.185	0.558	1.093	36.94	110.28
N	165	59	165	163	92

Notes: The table shows the estimated effects of a \$100,000 increase in government grants, adding a one-year lagged grant term to the specification, estimated separately by baseline shelter size. Size is defined by 2013 revenue terciles. Observations are at the shelter-fiscal-year level from 2013–2023. All regressions include shelter and year fixed effects, estimated separately within each tercile. Data come from the DV Shelter Database. Standard errors in parentheses are clustered at the shelter level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 33: Effect of Government Grants on Hospital Discharges by Shelter Size

	(1)	(2)	(3)
	IPV Discharge	DV Discharge	Assault Discharge
	Rate per 100k	Rate per 100k	Rate per 100k
<i>Panel A: Small Shelters</i>			
Grants (\$100k)	-0.490*	-0.577	-5.911
	(0.237)	(0.394)	(6.744)
Grants (lagged)	0.264	0.471	23.177**
	(0.357)	(0.599)	(9.903)
Sample Mean	5.751	11.974	637.894
N	90	90	90
<i>Panel B: Medium Shelters</i>			
Grants (\$100k)	-0.309**	-0.415	5.779
	(0.133)	(0.550)	(7.086)
Grants (lagged)	0.088	0.366	23.131**
	(0.309)	(0.754)	(9.875)
Sample Mean	6.395	13.520	515.152
N	90	90	90
<i>Panel C: Large Shelters</i>			
Grants (\$100k)	0.435*	0.390	-0.735
	(0.228)	(0.316)	(4.315)
Grants (lagged)	-0.381*	-0.295	2.015
	(0.212)	(0.360)	(4.377)
Sample Mean	5.853	12.735	431.064
N	84	84	84

Notes: The table shows the estimated effects of a \$100,000 increase in government grants on county-level hospital discharge rates, estimated separately by baseline shelter size. Shelter grant funding is aggregated to the county level as described in Section 4. Size is defined by 2013 revenue terciles. All regressions include county and year fixed effects, estimated separately within each tercile. Standard errors in parentheses are clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 34: Effect of Government Grants on Homicide Rates by Shelter Size

	(1)	(2)	(3)	(4)
	IPV Homicide	DV Homicide	Female Homicide	Child Homicide
<i>Panel A: Small Shelters</i>				
Grants (\$100k)	-0.057 (0.046)	0.250** (0.111)	-1.414** (0.550)	-0.044 (0.104)
Grants (lagged)	0.251* (0.142)	-0.047 (0.113)	-0.459 (0.800)	-0.030 (0.206)
Sample Mean	0.674	1.152	3.780	1.108
N	90	90	90	90
<i>Panel B: Medium Shelters</i>				
Grants (\$100k)	-0.170*** (0.050)	-0.216*** (0.068)	0.135 (0.191)	0.032 (0.148)
Grants (lagged)	0.162** (0.059)	0.220** (0.075)	-0.235 (0.235)	0.041 (0.176)
Sample Mean	0.710	1.012	2.652	1.510
N	90	90	90	90
<i>Panel C: Large Shelters</i>				
Grants (\$100k)	0.018 (0.026)	0.053 (0.033)	-0.058 (0.069)	-0.047 (0.132)
Grants (lagged)	-0.006 (0.021)	-0.041 (0.026)	0.216* (0.116)	-0.034 (0.143)
Sample Mean	0.460	0.842	1.605	1.755
N	84	84	84	84

Notes: The table shows the estimated effects of a \$100,000 increase in government grants on county-level homicide rates, estimated separately by baseline shelter size. Shelter grant funding is aggregated to the county level as described in Section 4. Size is defined by 2013 revenue terciles. All regressions include county and year fixed effects, estimated separately within each tercile. Standard errors in parentheses are clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 35: Effect of Government Grants on Child Maltreatment Reports by Shelter Size

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
All Ages	Infant	Ages 1–3	Ages 4–10	Ages 11–18	DV Risk	Foster Services	
Substantiated Reports							
Panel A: Small Shelters							
Grants (\$100k)	-4.816 (10.393)	-1.423 (3.016)	-1.991 (1.859)	2.430 (5.554)	-3.831 (2.892)	-7.162 (4.601)	0.339 (2.694)
Grants (lagged)	26.674 (48.959)	-2.741 (7.437)	10.391 (8.402)	8.233 (20.841)	10.791 (12.867)	14.180 (15.817)	4.197 (7.236)
Sample Mean	649.459	99.323	125.063	260.153	164.920	205.303	196.984
N	90	90	90	90	90	90	90
Panel B: Medium Shelters							
Grants (\$100k)	-21.853 (15.657)	-0.487 (2.505)	-2.965 (3.754)	-8.545 (6.831)	-9.856** (4.402)	-5.755 (4.399)	-3.581 (3.920)
Grants (lagged)	-6.271 (18.105)	1.563 (2.563)	-2.332 (3.579)	-1.869 (8.542)	-3.632 (4.624)	1.662 (5.777)	1.587 (4.039)
Sample Mean	654.386	108.175	126.805	257.861	161.545	245.978	182.981
N	90	90	90	90	90	90	90
Panel C: Large Shelters							
Grants (\$100k)	-13.595 (12.436)	-1.902 (1.185)	-2.496 (2.763)	-6.167 (6.062)	-3.030 (3.106)	0.958 (5.810)	0.374 (3.835)
Grants (lagged)	13.845 (11.217)	1.370 (1.075)	2.331 (2.417)	7.926 (5.474)	2.218 (3.207)	0.617 (7.219)	-2.592 (2.822)
Sample Mean	668.586	100.684	123.789	271.963	172.150	219.656	223.532
N	84	84	84	84	84	84	84

Notes: The table shows the estimated effects of a \$100,000 increase in government grants on county-level substantiated child maltreatment report rates, estimated separately by baseline shelter size. Shelter grant funding is aggregated to the county level as described in Section 4. Size is defined by 2013 revenue terciles. All regressions include county and year fixed effects, estimated separately within each tercile. Standard errors in parentheses are clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

F Additional Robustness

F.1 December Fiscal Year Fillings

Table 36 reports the full set of financial and staffing outcomes for the December fiscal year comparison summarized in Table 16.

Table 36: Robustness: December Fiscal Year Shelters, Full Results

	(1)	(2)	(3)	(4)	(5)
	Full Sample		December FY Only		Sample
	Coef.	SE	Coef.	SE	Mean
<i>Panel A: Revenue (\$100k)</i>					
Total Revenue	0.619***	(0.147)	0.867***	(0.179)	11.357
Private Donations	-0.389**	(0.161)	-0.086	(0.132)	2.692
Program Revenue	0.021	(0.022)	-0.007	(0.006)	0.209
Other Revenue	-0.013	(0.030)	-0.040	(0.051)	0.752
Grant Share	0.031***	(0.007)	0.023***	(0.008)	0.666
Private Donation Share	-0.027***	(0.007)	-0.013**	(0.006)	0.247
Program Revenue Share	-0.000	(0.001)	-0.001	(0.001)	0.018
Other Revenue Share	-0.003	(0.002)	-0.009**	(0.003)	0.069
<i>Panel B: Expenditure (\$100k)</i>					
Total Expenditure	0.573***	(0.069)	0.636***	(0.093)	10.500
Primary Program	0.447***	(0.063)	0.603***	(0.043)	8.242
Compensation	0.316***	(0.050)	0.355***	(0.066)	6.060
Contract Work	0.047*	(0.028)	-0.016	(0.027)	0.493
Operations	0.034	(0.024)	0.069*	(0.036)	0.323
Occupancy	-0.009	(0.021)	0.017*	(0.009)	0.621
Fundraising	0.051	(0.040)	0.083	(0.075)	0.461
Other	0.135**	(0.059)	0.128	(0.095)	2.542
<i>Panel C: Staffing</i>					
Employee Count	0.411*	(0.207)	0.621**	(0.251)	25.331
Volunteer Count	-1.319	(1.581)	-0.986	(1.136)	56.690
Observations	442		237		

Notes: Columns (1)–(2) reproduce the main TWFE estimates from Tables 7, 8, and 9, which include a lagged grant control (not shown). Columns (3)–(4) restrict the sample to the 24 shelters with December fiscal year ends within the 46-shelter analysis panel; one additional December-FY shelter is excluded from the analysis panel entirely, as described in Section 3. Highest paid employee compensation is omitted from this table, consistent with its exclusion from the main results, since only 6 shelters report it consistently. Sample means are from the December FY sample. All regressions include shelter and year fixed effects with standard errors clustered at the shelter level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Pre-2020 Sample

Table 37 restricts the financial sample to fiscal years 2013–2019, excluding the COVID-19 period and aligning the financial sample with the violence outcome data. Total revenue and total expenditure remain significant, though of somewhat different magnitude than the full-sample estimates: total revenue falls from \$61,900 in the main specification to \$50,000 pre-2020, while total expenditure rises from \$57,300 to \$66,100. Compensation, employee count, and operations expenditure remain significant and are also somewhat larger pre-2020.

The private donations coefficient is significantly negative in this subsample as well, and larger than the full-sample estimate ($-\$50,100$ per $\$100,000$ in grants, $p < 0.01$, vs. $-\$38,900$ in the main specification). The grant share and private donation share coefficients are also larger in magnitude pre-2020 (grant share 0.034 vs. 0.031 in the main specification; private donation share -0.031 vs. -0.027). This is consistent with the main specification’s crowd-out finding, not a departure from it: the pre-2020 subsample shows the same pattern, somewhat amplified.

The volunteer count coefficient is not significant in the pre-2020 sample, consistent with the main specification. Overall, the pre-2020 results indicate that the revenue, expenditure, compensation, and employee count findings are not artifacts of COVID-era disruptions to shelter operations or grant funding patterns; if anything, several estimates are somewhat larger in the pre-2020 period than in the full sample.

Table 37: Robustness: Pre-2020 Sample

	(1)	(2)	(3)	(4)	(5)
	Main Spec (2013–2023)		Pre-2020 (2013–2019)		Sample
	Coef.	SE	Coef.	SE	Mean
<i>Panel A: Revenue (\$100k)</i>					
Total Revenue	0.619***	(0.147)	0.500**	(0.200)	11.012
Private Donations	-0.389**	(0.161)	-0.501**	(0.195)	2.495
Program Revenue	0.021	(0.022)	0.013	(0.013)	0.248
Other Revenue	-0.013	(0.030)	-0.012	(0.050)	0.821
Grant Share	0.031***	(0.007)	0.034***	(0.007)	0.683
Private Donation Share	-0.027***	(0.007)	-0.031***	(0.008)	0.223
Program Revenue Share	-0.000	(0.001)	-0.000	(0.001)	0.024
Other Revenue Share	-0.003	(0.002)	-0.003	(0.003)	0.071
<i>Panel B: Expenditure (\$100k)</i>					
Total Expenditure	0.573***	(0.069)	0.661***	(0.127)	9.355
Primary Program	0.447***	(0.063)	0.540***	(0.082)	7.893
Compensation	0.316***	(0.050)	0.389***	(0.100)	6.197
Contract Work	0.047*	(0.028)	0.030**	(0.013)	0.484
Operations	0.034	(0.024)	0.059***	(0.020)	0.398
Occupancy	-0.009	(0.021)	0.007	(0.016)	0.586
Fundraising	0.051	(0.040)	0.030**	(0.012)	0.432
Other	0.135**	(0.059)	0.147***	(0.046)	2.407
<i>Panel C: Staffing</i>					
Employee Count	0.411*	(0.207)	0.569**	(0.254)	27.214
Volunteer Count	-1.319	(1.581)	-0.930	(2.570)	76.956
Observations	442		262		

Notes: Columns (1)–(2) reproduce the main TWFE estimates from Tables 7, 8, and 9. Columns (3)–(4) restrict the sample to fiscal years 2013–2019 to assess whether results are driven by COVID-19-era variation in shelter finances. Highest paid employee compensation is omitted from this table, consistent with its exclusion from the main results, since only 6 shelters report it consistently. Sample means are from the pre-2020 sample. All regressions include shelter and year fixed effects with standard errors clustered at the shelter level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

F.2 Sensitivity to Omitted Time-Varying Confounders

This appendix reports two checks bearing on the possibility that within-shelter changes in grant funding are correlated with unobserved, time-varying shelter or county characteristics that also affect outcomes. Section 4.2 presents the primary evidence on this concern: balancing regressions, an event study, and a direct examination of the grant allocation process. The checks reported here are complementary rather than primary, and should be read alongside that evidence rather than in isolation.

F.2.1 Time-Varying County Controls

Table 38 adds three time-varying county-level controls, median household income, poverty rate, and unemployment rate, to the main specification. Both specifications are estimated on the same 2013–2023 sample, so the comparison is not affected by any difference in sample composition. Coefficients are broadly stable across most outcomes: total revenue, total expenditure, compensation, and employee count all remain significant with controls added, with modest changes in magnitude in either direction. The private donation share coefficient remains significant and of similar magnitude with controls.

One result changes meaningfully with the addition of county controls: volunteer count, not significant in the main specification, becomes marginally significant and negative with controls added (-4.97 , $p = 0.072$). Given that this is the only outcome that changes significance status in either direction, I do not read this as evidence that the main specification is misspecified; the overall stability of the headline revenue, expenditure, and staffing results across the two specifications is consistent with the identifying variation coming primarily from within-shelter changes in grant funding rather than county-level economic conditions, though county-level income, poverty, and unemployment are a limited set of controls and do not rule out confounding from characteristics these variables do not capture.

Table 38: Robustness: Including Time-Varying County Controls

	(1)	(2)	(3)	(4)	(5)
	Main Spec		With County Controls		Sample
	Coef.	SE	Coef.	SE	Mean
<i>Panel A: Revenue (\$100k)</i>					
Total Revenue	0.619***	(0.147)	0.613***	(0.148)	12.888
Private Donations	-0.389**	(0.161)	-0.393**	(0.163)	3.046
Program Revenue	0.021	(0.022)	0.020	(0.021)	0.273
Other Revenue	-0.013	(0.030)	-0.014	(0.031)	0.857
Grant Share	0.031***	(0.007)	0.031***	(0.007)	0.680
Private Donation Share	-0.027***	(0.007)	-0.028***	(0.007)	0.226
Program Revenue Share	-0.000	(0.001)	-0.000	(0.001)	0.022
Other Revenue Share	-0.003	(0.002)	-0.003	(0.002)	0.072
<i>Panel B: Expenditure (\$100k)</i>					
Total Expenditure	0.573***	(0.069)	0.569***	(0.069)	11.989
Primary Program	0.447***	(0.063)	0.440***	(0.061)	9.081
Compensation	0.316***	(0.050)	0.315***	(0.050)	7.008
Contract Work	0.047*	(0.028)	0.051*	(0.028)	0.513
Operations	0.034	(0.024)	0.035	(0.024)	0.469
Occupancy	-0.009	(0.021)	-0.009	(0.021)	0.661
Fundraising	0.051	(0.040)	0.052	(0.040)	0.519
Other	0.135**	(0.059)	0.126**	(0.059)	2.820
<i>Panel C: Staffing</i>					
Employee Count	0.411*	(0.207)	0.390**	(0.171)	27.24
Volunteer Count	-1.319	(1.581)	-4.975*	(2.672)	53.98
Observations	442		442		

Notes: Columns (1)–(2) reproduce the main TWFE estimates from Tables 7, 8, and 9. Columns (3)–(4) add time-varying county-level controls: median household income, poverty rate, and unemployment rate. Both specifications are estimated on the same 2013–2023 sample; sample sizes are identical. Highest paid employee compensation is omitted from this table, consistent with its exclusion from the main results, since only 6 shelters report it consistently. Sample means are from the controls specification sample. All regressions include shelter and year fixed effects with standard errors clustered at the shelter level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

F.2.2 Oster Bounding

I additionally assess sensitivity to selection on unobservables following Oster (2019), which uses the movement in coefficients and R^2 between an uncontrolled specification (year fixed effects only) and the controlled specification (shelter and year fixed effects) to bound the degree of selection on unobservables required to explain away an estimated effect. The method produces δ , the ratio of selection on unobservables to selection on observables that would be required to attribute the full effect to omitted variables. A common benchmark, following **oster2019**<empty citation>, treats $|\delta| \geq 1$ as evidence that an effect is not easily explained by omitted confounders, since it would require unobservables to matter at least as much as the observables already controlled for.

Table 39 reports δ for the five outcomes examined. Four outcomes clear the $|\delta| \geq 1$ benchmark: Total Revenue ($\delta = 1.53$), Total Expenditure ($\delta = 3.24$), Compensation ($\delta = 1.98$), and Employee Count ($\delta = 1.90$). For these outcomes, unobserved selection would need to be at least as strong as the selection already absorbed by shelter fixed effects to account for the estimated effect, and in three cases substantially stronger.

Grant Share's δ estimate (-10.52) is both sign-reversed and far outside the range in which this statistic is directly interpretable. This pattern arises when the change in R^2 between the uncontrolled and controlled specifications is large relative to the change in the coefficient, which mechanically produces an extreme or sign-reversed δ regardless of the underlying stability of the estimate. In this case, I instead report the coefficient identified under the conventional $\delta = 1$ benchmark: 0.036, close to the controlled specification's own estimate of 0.031. This indicates that Grant Share's estimate is not particularly sensitive to a plausible degree of selection on unobservables, even though the δ statistic itself is uninformative here.

Across all five outcomes, I do not find evidence that selection on unobserved shelter characteristics is likely to explain the estimated effects. As with any Oster bounding exercise, this is one piece of evidence to be weighed alongside the identification checks in Section 4.2

— particularly the direct examination of the grant allocation process — rather than read as a formal test.

Table 39: Oster Bounding: Selection on Unobservables

Outcome	Uncontrolled $\hat{\beta}$	Controlled $\hat{\beta}$	Controlled R^2	δ
Total Revenue	0.950	0.619	0.942	1.53
Total Expenditure	0.927	0.573	0.977	3.24
Compensation	0.529	0.316	0.948	1.98
Employee Count	0.81	0.38	0.856	1.90
Grant Share	0.020	0.031	0.759	−10.52

Notes: Uncontrolled specification includes year fixed effects only; controlled specification includes shelter and year fixed effects, matching the main specification. δ is the Oster (2019) bounding parameter, with R_{max} set to 1.3 times the controlled R^2 (capped at 1), following convention. Revenue, expenditure, and compensation coefficients are expressed in \$100,000 units. Grant Share’s δ is sign-reversed and uninformative; the coefficient identified at the conventional $\delta = 1$ benchmark is 0.036, close to the controlled estimate reported here. Outcomes with an already-insignificant coefficient in the main specification (private donations, program revenue, other revenue, volunteer count) are omitted from this table; Oster bounds are less informative for effects not distinguishable from zero.